



Design and Mathematical Modelling of a Grid-Monitoring Remote Terminal Unit for SCADA in Wind Farms

University of Portsmouth — School of Energy and Electrical Engineering

Renewable Energy Engineering - BEng (Hons) Final Year Dissertation

Jack Davies — Student ID 2246819

Project Supervisor: Dr Lei Kang

Ethics Approval: TETHIC-2025-112031

Submitted May 2026

Abstract

The integration of multi-megawatt wind generation into the GB transmission network requires accurate real-time monitoring of electrical parameters at the point of common coupling so that the system operator can verify continuous compliance with the GB Grid Code, EREC G5/5 and IEC 61400-21. Commercial Remote Terminal Units (RTUs) are mature products supplied by major OEMs, but they are closed systems whose internal mathematical models, firmware architectures and signal-conditioning topologies are proprietary. This presents an obstacle to undergraduate engineers who need to understand the measurement chain end to end. This dissertation presents the design, analytical modelling, firmware implementation and laboratory validation of a low-cost (< £100), open mathematical model RTU proxy targeted at the post-converter grid-side signals of a direct-drive PMSG wind turbine (Siemens Gamesa SG 14-222 DD).

The system is formally classified as active, open-loop, non-linear, time-invariant, discrete and MIMO. A six-element active signal-conditioning circuit — comprising a loaded voltage divider, AC coupling, a 1.65 V DC bias network, two MCP6002 unity-gain buffers and a first-order anti-aliasing low-pass stage — is derived from Kirchhoff's voltage and current laws, characterized in the Laplace domain, and confirmed analytically and in LTspice to be asymptotically stable while preserving the inter-phase 120-degree relationship necessary for constant three-phase power.

Seven firmware modules (M0 through M6) implement ADC bias calibration, RMS, frequency and Rate of Change of Frequency (RoCoF), phase angle and power factor, three-phase active/reactive/apparent power, harmonic analysis via a direct 40-point DFT, and DNP3 (IEEE 1815) Group 30 Variation 5 and Group 1 Variation 1 telemetry encoding under the IEC 61400-25 WGEN/WCNV/GGIO logical-node mapping.

The SISO build was validated on solderless breadboard with the MCP6002 op-amp and subsequently transferred to a permanent PC01603 stripboard; all seven modules produced results either within the IEC 61400-21 Class I $\pm 0.5\%$ accuracy band or were documented against their PDS targets with proposed resolutions. The build was then expanded to a three-phase MIMO architecture using a dual-Feather Strategy B in which Feather 1 generates V_c and measures $V_a/V_b/V_c/I_b/I_c$ while Feather 2 generates V_a and V_b and measures I_a , with a GPIO27 hardware sync line resetting Feather 2's DAC index on every phase measurement. The MIMO build has been validated both on solderless breadboard and on a separate stripboard, with all six channels live, all alarms behaving correctly across three disturbance conditions, and DNP3 frames passing CRC verification on every transmission.

The hardware human-machine interface tier passed through two failed routes — a Grove SH1107 I²C OLED and a KMRTM28028 ILI9341 SPI TFT — both of which crashed the integrated firmware loop because their bus transactions contended with the 500 ms DNP3 transmit cycle on the same active core. The final deliverable replaces the on-device display with a Python SCADA HMI dashboard running on the host laptop, consuming the DNP3 byte stream over USB-serial, decoding the Group 30 / Group 1 objects, and rendering twelve live analogue parameters, six alarm tiles, a PDS compliance summary strip and a live frame inspector. This relocation of the HMI from the device tier to the host tier is consistent with the SCADA Level 3 (CAN) position described in Chapter 2 and is the architecturally correct response to the on-device display contention rather than a workaround.

The work shows that an educationally accessible, fully documented RTU proxy can satisfy the principal GB Grid Code monitoring requirements and provides a reproducible reference for future student projects in grid-side power-quality instrumentation.

Keywords: Remote Terminal Unit (RTU), wind energy, GB Grid Code, IEC 61400-21, EREC G5/5, DNP3 (IEEE 1815), SCADA, ESP32, signal conditioning, Bode analysis, power quality, three-phase, LCL filter, Python SCADA HMI.

Declaration. This dissertation is my own work. Any material derived from other sources has been duly acknowledged. Ethics approval for the project was granted under reference TETHIC-2025-112031 on 4 November 2025.

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List of Abbreviations

Abbreviation	Expansion
ADC	Analogue-to-Digital Converter
AC	Alternating Current
BoM	Bill of Materials
CAN	Control Area Network (SCADA Level 3)
CRC	Cyclic Redundancy Check
DAC	Digital-to-Analogue Converter
DC	Direct Current
DFT	Discrete Fourier Transform
DNL	Differential Non-Linearity
DNP3	Distributed Network Protocol 3 (IEEE Std 1815)
EMI	Electromagnetic Interference
EREC	Engineering Recommendation (UK distribution-network engineering)
FFT	Fast Fourier Transform
FAN	Farm Area Network (SCADA Level 2)
GBW	Gain Bandwidth Product
GGIO	Generic Grid I/O (IEC 61400-25 logical node)
HMI	Human-Machine Interface
IEC	International Electrotechnical Commission
IGBT	Insulated-Gate Bipolar Transistor
LCL	Inductor–Capacitor–Inductor (filter topology)
LSB	Least Significant Bit
LTI	Linear Time-Invariant
MIMO	Multiple-Input Multiple-Output
MMS	Manufacturing Message Specification (IEC 61850-8-1)
NESO	National Energy System Operator (Great Britain)
OLED	Organic Light-Emitting Diode
PCC	Point of Common Coupling
PDS	Product Design Specification
PF	Power Factor
PMSG	Permanent-Magnet Synchronous Generator
PT / CT	Potential Transformer / Current Transformer
RC	Resistor–Capacitor
RoCoF	Rate of Change of Frequency

Abbreviation	Expansion
RMS	Root Mean Square
RTU	Remote Terminal Unit
SAR	Successive-Approximation Register (ADC architecture)
SCADA	Supervisory Control and Data Acquisition
SISO	Single-Input Single-Output
SPI	Serial Peripheral Interface
TAN	Turbine Area Network (SCADA Level 1)
TFT	Thin-Film Transistor
THD	Total Harmonic Distortion
WCNV	Wind Converter (IEC 61400-25 logical node)
WGEN	Wind Generator (IEC 61400-25 logical node)
WTC	Wind Turbine Controller

Chapter 1 — Introduction

1.1 Background and Motivation

Global wind capacity and its UK contribution have grown sharply over the last decade and continue to do so under the 2050 net-zero target embedded in the Climate Change Act 2008 [1], [2]; the market context is treated in detail in Section 2.1. What follows from that growth is a practical engineering requirement: the inherent intermittency of wind, characterized by ramp events, frequency deviations and voltage transients, demands continuous, high-resolution data acquisition at the Point of Common Coupling (PCC) if the system is to remain stable under variable generation [3].

The field device that performs that acquisition and forwards the data to the supervisory control hierarchy is the Remote Terminal Unit (RTU). The global market for RTUs in smart-grid applications was valued at approximately \$2.73 billion in 2025 and is forecast to reach \$3.04 billion by 2030 at a compound annual growth rate of 11.6 % [4]. Medium-voltage units make up the largest market share, driven by their cost-effectiveness in substation and renewable-energy applications, and adoption of open communication protocols — principally DNP3 (IEEE 1815) and IEC 61850 — has continued to grow as operators seek interoperability across heterogeneous SCADA platforms [5].

Commercial RTUs from the major OEMs (Siemens, ABB, GE Vernova) are mature, capable and widely deployed in both offshore and onshore wind farms [5]. They are also closed systems: their internal mathematical models, firmware architectures and signal-conditioning topologies are proprietary. For an undergraduate or research engineer attempting to understand the complete chain from sensor through algorithm to telemetry frame, the lack of an openly documented reference design is a significant pedagogical gap. The same gap is visible in the published

academic literature, which tends to focus either on high-level SCADA architecture [6] or on advanced control algorithms, with the practical signal-acquisition chain rarely treated in detail.

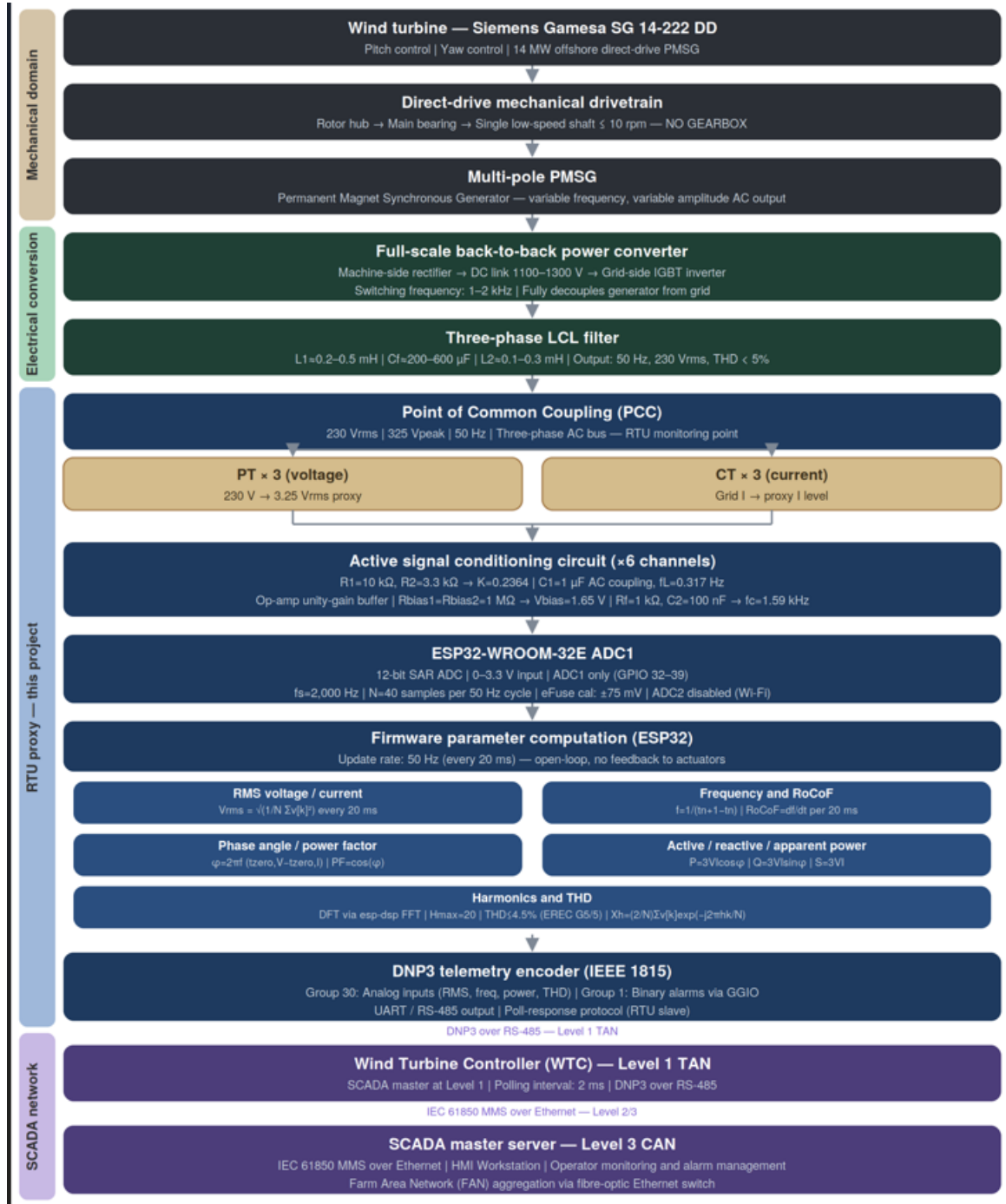


Figure 1.1 - Wider system block diagram of the grid-monitoring RTU proxy.

1.2 Problem Statement

The intermittent character of wind generation makes continuous high-resolution monitoring at the PCC essential for grid stability. Commercially available RTUs satisfy this requirement but are

expensive, closed and not easily adaptable to laboratory-scale teaching or research work. There is an identified gap for an accessible, lab-based proxy RTU design that demonstrates the complete measurement chain — analogue signal conditioning, discrete-time mathematical modelling, embedded firmware implementation and standards-conformant telemetry — while remaining compliant with UK statutory limits, faithfully reproducible by a third party and inexpensive to construct.

1.3 Aim and Objectives

The aim of the project is to design, model, build and laboratory-validate a low-cost RTU proxy that acquires post-converter three-phase voltage and current signals, derives the GB Grid Code compliance parameters from them, and telemeters those parameters to a SCADA master via the DNP3 protocol. The aim is decomposed into four objectives.

- **Objective 1 - Theoretical and Technical Foundation.** Establish a structured knowledge base covering wind-energy conversion, the SCADA hierarchy, the LCL filter that produces grid-compliant post-converter signals, and the UK statutory limits (GB Grid Code, EREC G5/5, IEC 61400-21). Include ethical considerations for sustainable engineering practice and identify the research gaps that the project addresses.
- **Objective 2 - Mathematical Model.** Derive the active signal-conditioning circuit's transfer function from Kirchhoff's voltage and current laws; classify its poles and confirm asymptotic stability; derive the seven discrete-time parameter-extraction algorithms (RMS, frequency, RoCoF, phase angle, power factor, three-phase power, THD) with explicit disturbance terms; and validate the model independently in both Python and LTspice.
- **Objective 3 - Firmware Implementation.** Implement the seven-module firmware on an Adafruit Feather ESP32-WROOM-32E (which carries the Espressif ESP32-WROOM-32E module) for the SISO development phase; expand to a six-channel three-phase MIMO architecture using a dual-Feather configuration; verify each module on a synthetic bench against a four-condition disturbance set; and verify on hardware against Aim TTi TG330 inputs and dual-Feather DAC-generated three-phase signals.
- **Objective 4 - Testing, SCADA Integration and Documentation.** Conduct structured disturbance testing under undervoltage sag, frequency deviation, noise and harmonic injection; integrate with a SCADA/HMI tier; and produce a final reproducible engineering report and risk assessment.

1.4 Scope, Assumptions and Limitations

The project develops a proxy RTU at laboratory scale. No high-voltage equipment, no rotating machinery and no field deployment are involved. All signal sources are generated at the bench: voltages by the Aim TTi TG330 function generator and the ESP32 on-chip DAC; currents are emulated by an RC phase-shift network on a parallel ADC channel rather than by a physical current transformer. The proxy uses the ESP32 internal 12-bit SAR ADC rather than a calibrated external sigma-delta ADC; ADC differential non-linearity and the 8-bit DAC staircase together set a measurable harmonic floor that is identified, characterized and addressed by the passive remediation framework of Section 3.5. The £100 component budget excludes a real-time operating system and external high-resolution ADCs but is sufficient to demonstrate the full algorithm chain.

1.5 Significance and Original Contribution

The principal original contribution of this dissertation is not a new algorithm or a new measurement technique. It is a fully documented, reproducible BEng-level reference RTU design with four properties not jointly available in either academic or commercial literature:

- (i) the active signal-conditioning circuit topology is derived stage-by-stage from Kirchhoff's laws, and its stability is formally verified;
- (ii) the seven parameter algorithms are mapped one-for-one to firmware modules with explicit equation cross-references and are unit-tested both synthetically and on hardware;
- (iii) the complete laboratory build is documented including its failures — the LM358 common-mode-range incident, the ESP32 ADC2 / Wi-Fi contention, the dual-Feather crystal-oscillator synchronization limitation that motivated the GPIO27 sync line, the two failed on-device HMI routes — and the engineering decisions made to resolve each one;
- (iv) the choice to relocate the HMI tier off the device and onto a Python host consuming the DNP3 stream is shown to be the correct architectural response rather than a workaround.

The work is intended as a teaching artefact as much as a research deliverable.

1.6 Report Structure

Chapter 2 establishes the theoretical and technical foundation: wind energy conversion, the SCADA hierarchy, the LCL filter, the UK compliance limits, the three-phase balanced-power property and the ethical context. Chapter 3 derives the discrete-time mathematical model, the conditioning-circuit transfer function, the pole/stability analysis and the seven discrete-time algorithms and validates the model through Python and LTspice simulation against four disturbance conditions. Chapter 4 reports the firmware implementation, the synthetic and hardware module validation, the integrated DNP3 disturbance testing, the MIMO three-phase expansion on breadboard and stripboard, and the Python SCADA HMI integration. Chapter 5 summarizes the findings, evaluates the work against the four objectives, and lists recommendations and future work. References and appendices (risk assessment, Python simulation, firmware listings, project planning timeline) follow.

Chapter 2 — Literature Review and Technical Foundation

This chapter establishes the theoretical and technical knowledge base required for the grid-monitoring RTU proxy. It follows the complete energy conversion pathway within a variable-speed wind turbine — from aerodynamic capture through to the conditioned electrical signals at the PCC — before turning to the SCADA architecture within which the RTU operates and the UK grid compliance standards that govern its measurement requirements. A top-down approach is used so that every design decision taken in later chapters is grounded in the physical and regulatory context of the system.

2.1 Market Context and Drivers

The global transition to low-carbon electricity has driven unprecedented growth in wind energy deployment. Cumulative installed wind capacity has risen from approximately 487 GW in 2015 to over 1 TW by the end of 2025 [1], and the legally binding 2050 net-zero target [2] ensures continued capital investment in offshore and onshore wind infrastructure within the United Kingdom. The RTU market within the smart-grid sector reflects the resulting demand: the global market was valued at approximately \$2.73 billion in 2025 and is forecast to reach \$3.04 billion by 2030 at a compound annual growth rate of 11.6 % [4]. Medium-voltage RTUs account for the largest share owing to their cost-effectiveness in substation and renewable-energy applications, and adoption of open communication protocols (principally DNP3 and IEC 61850) continues to drive interoperability across heterogeneous SCADA platforms [5]. Emerging applications — including AI-assisted predictive maintenance and adaptive fault detection — further reinforce commercial demand for scalable, low-cost RTU architectures [7]. This project addresses an identified gap: accessible, laboratory-scale proxy RTU designs accompanied by open mathematical models and firmware listings are scarce in the published literature, yet they are essential for educating the next generation of power-systems engineers.

2.2 Energy Transfer and Conversion in Wind Turbines

A direct-drive variable-speed wind turbine converts kinetic energy in the wind into grid-compliant electrical power through a sequential, multi-stage process. Understanding each stage is essential for justifying why the RTU is positioned at the grid side of the power converter and why post-conversion signals can be treated as generator-independent for monitoring purposes.

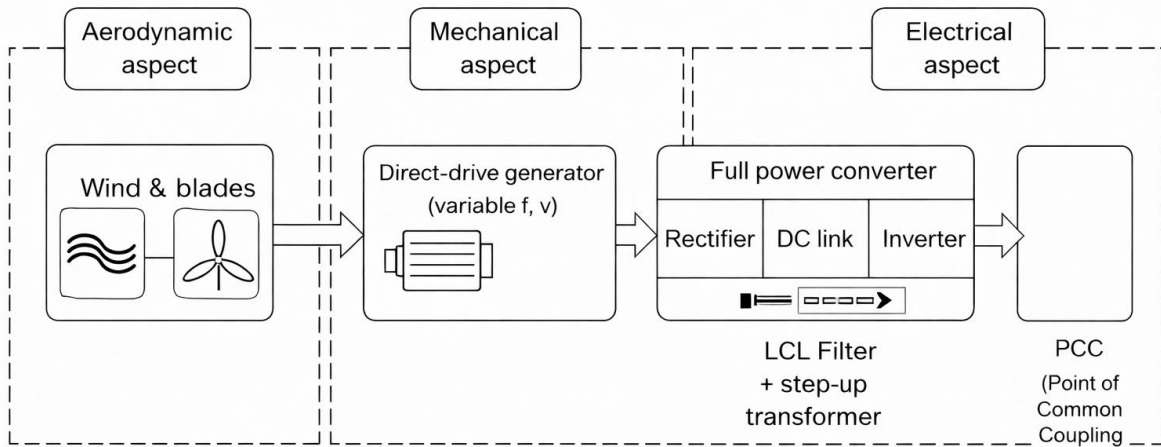


Figure 2.1 - Energy transfer pathway in a direct-drive wind turbine.

2.2.1 Aerodynamic Capture

Aerodynamic conversion begins when the wind applies lift and drag forces to the rotor blades. The maximum theoretical fraction of the wind's kinetic energy that any rotor can extract is bounded by Betz's limit, $C_{p,max}=16/27 \approx 0.593$ [8], [9]:

$$P_{aero} = \frac{1}{2} \rho A C_p v^3 \quad (2.1)$$

where ρ is air density (1.225 kg/m³ at sea level), A is the swept rotor area, C_p is the power coefficient (a function of tip-speed ratio and pitch angle), and v is the free-stream wind speed. Modern utility-scale rotors achieve operational C_p values in the range 0.45–0.50, somewhat below the theoretical maximum because of finite blade count, tip losses and viscous drag [10].

The reference platform for this project is the Siemens Gamesa SG 14-222 DD direct-drive offshore turbine [11], with a 222 m rotor diameter, a rated power of 14 MW and a swept area of approximately 38 700 m². Direct-drive topology eliminates the gearbox and connects the rotor directly to a large multi-pole Permanent Magnet Synchronous Generator (PMSG), which produces variable-frequency, variable-amplitude AC that is decoupled from the grid by a full-scale back-to-back power converter.

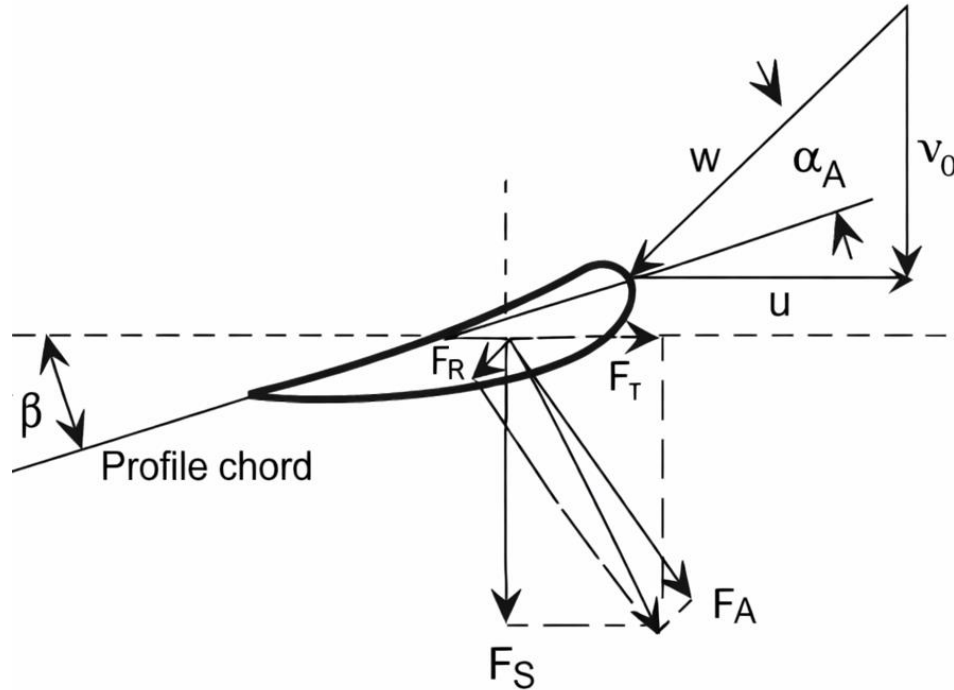


Figure 2.2 - Velocity triangle and aerodynamic forces on a blade section.

2.2.2 Mechanical and Generator Conversion

The PMSG converts the rotor's mechanical shaft power to electrical power at a frequency proportional to rotor speed and pole pair count. For the SG 14-222 DD operating between cut-in and rated wind speeds the generator's electrical frequency varies continuously, making its output incompatible with a fixed-frequency grid without intermediate conversion. The full-scale back-to-

back converter rectifies the PMSG output to a DC link and re-inverts it at 50 Hz, completely decoupling the generator electrical characteristics from the grid-side terminals.

The grid-side characteristics seen by the RTU therefore depend only on the converter and its LCL filter, not on the generator, and the measurement chain is generator-independent.

2.3 The LCL Filter at the Converter Output

The full-scale IGBT inverter at the grid side of the converter switches at 1–10 kHz and would inject substantial high-order harmonic content into the grid without filtering. The LCL filter (inverter-side inductance L_1 , shunt capacitor C_f with damping resistor R_d , grid-side inductance L_2) is the standard topology for this attenuation [12], [13]. Typical multi-megawatt values are $L_1 \approx 0.2\text{--}0.5$ mH, $C_f \approx 200\text{--}600$ μF , $R_d \approx 1\text{--}10$ Ω and $L_2 \approx 0.1\text{--}0.3$ mH; for the LTspice simulation presented in Chapter 3, the values are scaled down to $L_1 = 2$ mH, $C_f = 10$ μF , $R_d = 10$ Ω , $L_2 = 1$ mH while preserving the topology and the 50 Hz operating point.

The LCL filter's resonance frequency, given by

$$f_{res} = \frac{1}{2\pi} \sqrt{((L_1 + L_2)/(L_1 \times L_2 \times C_f))} \quad (2.2)$$

evaluates for the scaled values to $f_{res} \approx 1.94$ kHz, sitting between ten times the 50 Hz grid frequency and one half of the IGBT switching frequency as recommended by Liserre et al. [12]. The damping resistor R_d in series with C_f suppresses the LC resonance peak; in commercial deployments this is replaced by active damping in the converter control loop, but for the simulation a passive resistor is sufficient.

The LCL output is the signal the RTU must monitor. In a full-scale deployment this is typically reduced to $3.25 V_{RMS}$ by a 1:100 potential transformer (PT) before reaching the RTU's signal-conditioning circuit; the bench TG330 function generator produces this $3.25 V_{RMS}$ proxy directly, bypassing the very large and high-voltage physical PT.

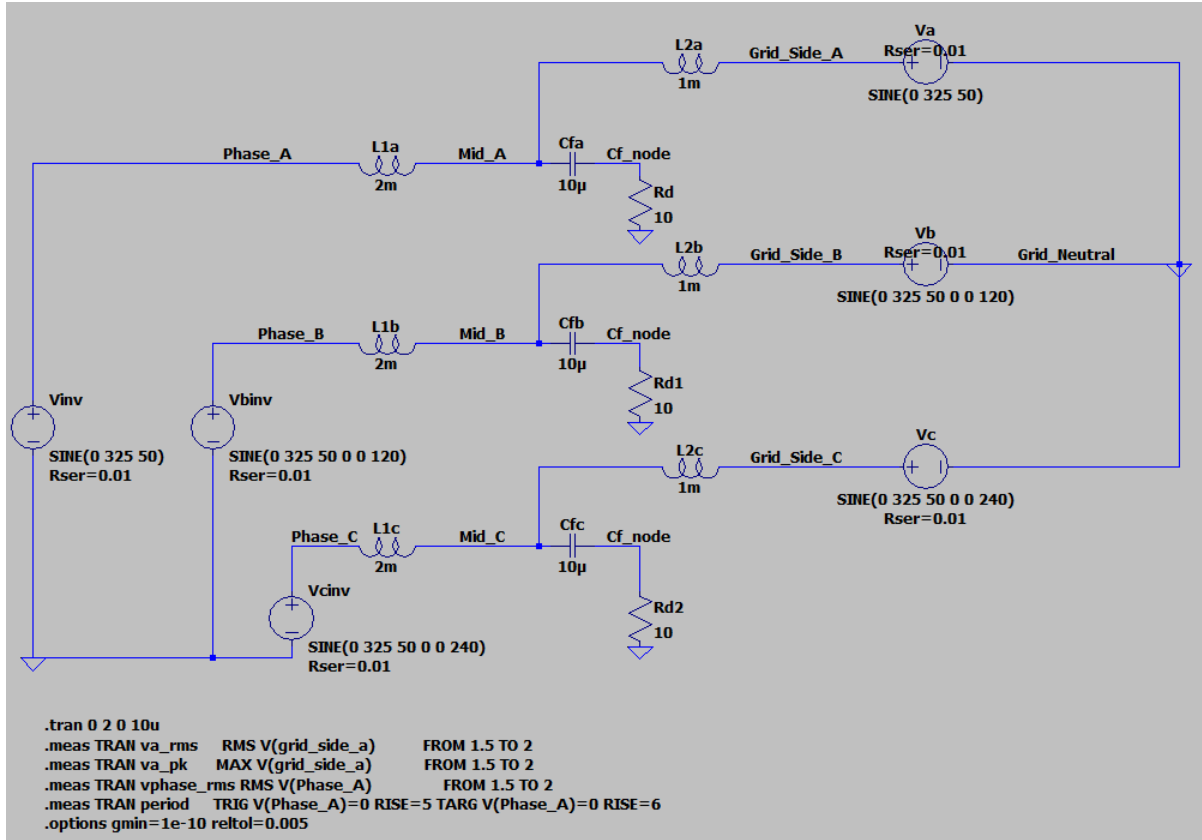


Figure 2.3 - LTspice schematic of the three-phase LCL filter.

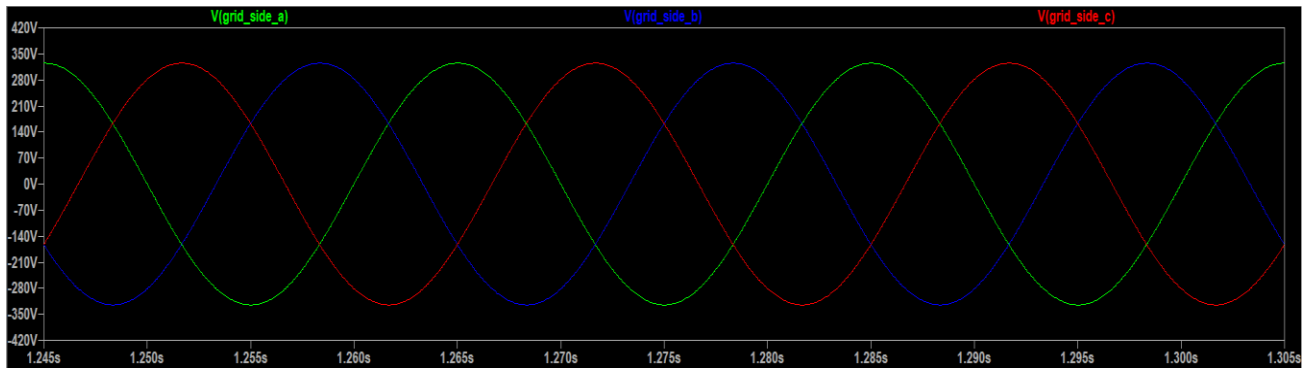


Figure 2.4 - Simulated three-phase voltage waveforms at the LCL output.

2.4 SCADA Architecture and the RTU's Position

The SCADA architecture of a modern wind farm is a three-level hierarchy summarized in Figure 2.5 [6]:

SCADA Hierarchy Diagram

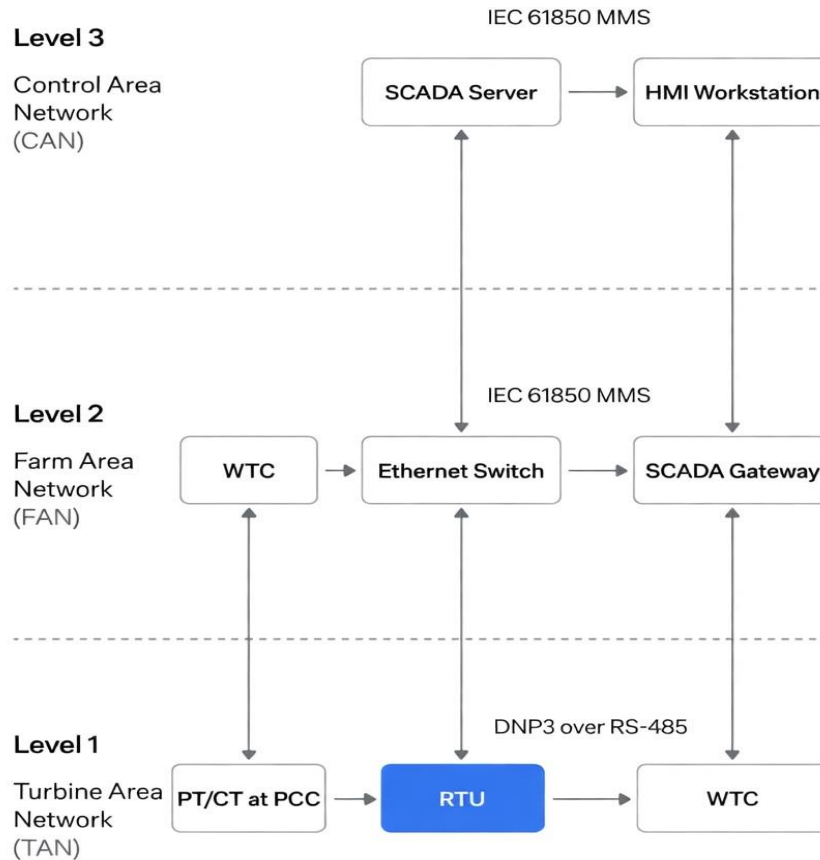


Figure 2.5 - Three-level SCADA hierarchy for a wind farm.

The RTU operates as a slave device at the lowest data-acquisition tier within this architecture. It responds to poll requests from the SCADA master (typically the WTC at Level 1) with a polling interval of 2 ms for critical electrical parameters [16]. Bidirectional control commands from the SCADA system to the turbine actuators (pitch drive, converter control) are processed by the WTC and not by the RTU. This separation of monitoring and control functions is a deliberate engineering simplification: restricting the RTU to data acquisition and telemetry only reduces cost, eliminates feedback-loop stability risk, and minimizes the attack surface for cybersecurity threats. The same separation makes the present open-loop classification of the RTU (Section 3.1) natural rather than a forced choice.

2.4.1 IEC 61400-25 Logical Nodes

IEC 61400-25 is the international standard governing information models and communication for wind power plants [15]. It defines a set of logical nodes that the RTU populates with measured parameters and alarm states:

- **WGEN (Wind Generator)** - reports terminal voltages, phase currents, active and reactive power, and frequency.
- **WCNV (Wind Converter)** - reports converter-related parameters including THD, switching state and DC-link voltage where instrumented.
- **GGIO (Generic Grid I/O)** - carries binary alarm states (undervoltage, over-frequency, imbalance, harmonic exceedance) derived from threshold comparisons within the RTU firmware.

The mandatory analogue-measured parameters required by IEC 61400-25 and cross-referenced against the GB Grid Code are three-phase RMS voltages, three-phase RMS currents, three-phase active and reactive power, apparent power, power factor, fundamental frequency, Rate of Change of Frequency, voltage-to-current phase angle per phase, and total harmonic distortion up to the 50th order. All of these are derived from the six scaled proxy signals (Va, Vb, Vc, Ia, Ib, Ic) acquired by the RTU ADC — no additional field instrumentation is required, which is a key economy justified by the generator-independent, post-converter monitoring philosophy of Section 2.2.

2.4.2 Communication Protocol Choice

Communication between the RTU and its master is implemented using industry-standard protocols. DNP3 (Distributed Network Protocol 3, IEEE Std 1815-2012 [14]) is the preferred protocol for this project. Three reasons justify the choice over IEC 61850-8-1 MMS: DNP3 has widespread adoption in UK distribution automation, its framing structure is simpler and its overhead lower (a Group 30 Variation 5 frame carrying ten analogue floats fits in 49 bytes versus the several-kilobyte overhead of an MMS connection), and it supports native timestamped event reporting and robust data integrity via CRC-16 with a domain-specific polynomial (0xA6BC). IEC 61850-8-1 MMS provides a complementary option for substation-level interoperability and is noted as a future upgrade pathway once the core proxy is validated.

2.5 UK Grid Compliance Standards and Measurement Methods

Grid compliance for wind energy in the United Kingdom is governed by the National Energy System Operator (NESO) GB Grid Code [17], supported by Engineering Recommendation EREC G99 (connection of generating plant ≥ 1 MW) and EREC G5/5 (harmonic voltage distortion planning levels) [18]. These documents impose quantitative limits on the electrical parameters that the RTU must measure and report in real time. Table 2.1 summarizes the key compliance limits relevant to this project.

Parameter	GB Grid Code Limit	Reference Clause	RTU Measurement Method
Frequency	49.5–50.5 Hz continuous; 47.5–52 Hz extended	CC6.1.2	Zero-crossing detection
Voltage (phase-to-neutral)	± 10 % of nominal (230 V)	CC6.1.4	RMS computation per cycle
Power factor	≥ 0.95 lagging/leading at full output	CC6.1.3	Phase-angle V–I measurement

Parameter	GB Grid Code Limit	Reference Clause	RTU Measurement Method
THD voltage (< 25 kV)	$\leq 4.5\%$ (planning level)	EREC G5/5 Table 1	DFT up to 50th harmonic
THD voltage (25–66 kV)	$\leq 3.7\%$ (planning level)	EREC G5/5 Table 1	DFT up to 50th harmonic
RoCoF withstand	≥ 1 Hz/s	CC6.1.2	Successive frequency estimates
Voltage imbalance	$< 1\%$ negative sequence	IEC 61400-21	Symmetrical-component analysis

Table 2.1 — Key GB Grid Code and EREC G5/5 compliance limits relevant to the RTU proxy, with the corresponding measurement methods implemented in the firmware. Reference clauses are taken from the GB Grid Code [17], ENA EREC G5/5 [18] and BS EN IEC 61400-21-1:2019+A11:2020 [19].

Each limit in Table 2.1 maps directly to a firmware algorithm validated in Chapter 4. The measurement principles for the most critical parameters are summarized below.

Frequency measurement — zero-crossing detection.

Frequency is the most fundamental grid compliance parameter; deviations outside 49.5–50.5 Hz indicate an imbalance between generation and demand that can cascade to system collapse. The RTU derives frequency by timestamping consecutive positive-going zero-crossings of the voltage waveform:

$$T = t_{\{n+1\}} - t_n, f = 1/T \quad (2.3)$$

, where successive zero-crossings are timestamped to microsecond precision using the ESP32 hardware timer. A moving-average filter applied over five consecutive period estimates suppress noise-induced jitter, achieving measurement uncertainty well within the ± 10 mHz target. This method is computationally inexpensive, requires no FFT processing, and provides a new frequency estimate every 20 ms — one per fundamental cycle — which is sufficient for both steady-state compliance monitoring and RoCoF detection [17].

RMS voltage and current.

Root-mean-square values quantify the effective power-delivering magnitude of the AC waveforms and are the primary indicators of over- and under-voltage events. For a discretized waveform sampled at rate f_s with N samples per cycle the discrete RMS estimator is

$$V_{rms} = \sqrt{\frac{1}{N} \sum_{n=0}^{N-1} v_a^2[k]} \quad (2.4)$$

This is computed once per fundamental cycle (every 20 ms) and threshold-compared against the $\pm 10\%$ GB Grid Code limit; any exceedance triggers a binary alarm event telemetered to SCADA via the GGIO logical node.

Rate of Change of Frequency (RoCoF).

RoCoF is a critical indicator of large generation–demand imbalances and is monitored by loss-of-mains protection schemes. It is computed as

$$RoCoF = \frac{df}{dt} = \frac{f[n] - f[n-1]}{\Delta t_{update}} \quad (2.5)$$

, where $\Delta t_{update} = 20$ ms for one-cycle averaging. The GB Grid Code requires generating plants to remain connected for RoCoF events up to 1 Hz/s; the RTU reports RoCoF continuously so that the SCADA system can issue a tripping command if this threshold is exceeded [17].

Total Harmonic Distortion (THD).

THD quantifies the degree to which the voltage or current waveform deviates from a pure sinusoid because of harmonic content from power-electronic switching. It is computed using a Discrete Fourier Transform (DFT) applied to one or more cycles of the sampled waveform.

$$V_h = \left(\frac{2}{N} \right) \left| \sum v[n] e^{(-\frac{j2\pi hn}{N})} \right|, h = 1, 2, \dots, 50 \quad (2.6)$$

$$THD = \frac{(\sum v_h^2 \text{ for } h=2 \text{ to } 50)}{V_1} \times 100\% \quad (2.7)$$

Per IEC 61400-21 and EREC G5/5 the THD of the voltage at the PCC must remain below the planning levels of Table 2.1, accounting for background harmonic levels already present on the network [18]. The RTU reports individual harmonic magnitudes up to the 50th order (2.5 kHz at full sample-rate) to the SCADA system, enabling the network operator to identify the source and order of any harmonic problem.

Phase angle and power factor.

The phase angle between voltage and current at the PCC determines the power factor and the reactive power exchange with the network. It is measured as the time delay between corresponding zero-crossings of the voltage and current waveforms on the same phase:

$$\varphi = 2\pi f_0 (t_{zero,V} - t_{zero,I}) \quad (2.8)$$

, with $PF = \cos \varphi$. The GB Grid Code requires a power factor of at least 0.95 (lagging or leading) at full output. Deviations indicate excessive reactive power consumption or generation, which may cause voltage instability at the PCC — a condition the RTU detects and reports immediately.

2.6 Three-Phase Power and the Phase-Balance Requirement

In a balanced three-phase system the instantaneous power on each individual phase pulsates at twice the fundamental frequency:

$$p_a(t) = V_{pk} \times I_{pk} \times \cos(2\omega t - \varphi) \quad (2.9)$$

The phase-displaced equivalents on phases b and c, when summed, yield a constant total power

$$P_{total} = 3V_{rms} \times I_{rms} \times \cos(\varphi) \quad (2.10)$$

This constant-power property depends critically on the 120° inter-phase separation being preserved through the entire measurement chain. The conditioning-circuit phase analysis of Section 3.5 confirms that this property is preserved in the present design — every channel attenuates and phase-shifts the signal identically, so the inter-channel relationship is invariant under the filter.

The symmetrical-component decomposition of Fortescue [40] resolves an unbalanced three-phase set into positive-, negative- and zero-sequence components; the negative-sequence component is the IEC 61400-21 voltage-imbalance metric of Table 2.1. The RTU therefore reports both raw per-phase RMS values (for over-/under-voltage detection) and the implicit balance information (via the three values that the SCADA master can post-process into negative-sequence magnitude). Full M7-level symmetrical component decomposition on-device is identified in Chapter 5 as future work.

2.7 Identified Research Gaps and Ethical Context

Three gaps in the existing literature are addressed by this work.

First, openly published BEng-level reference designs for grid-monitoring RTUs are scarce; commercial designs are proprietary and academic papers typically address SCADA architecture [6] or advanced control rather than the practical signal-acquisition chain. Where reference designs do exist, they tend to omit either the active conditioning topology or the discrete-time algorithm derivation.

Second, detailed block diagrams linking the complete wind-turbine energy conversion pathway to the RTU monitoring point — showing the full signal flow from rotor to ADC input — are similarly rare, leaving students with little guidance on the physical basis of the measurement system. Figure 1.1 of this report addresses the gap directly.

Third, the active signal-conditioning circuit topology used here — incorporating both an AC-coupling stage and a downstream anti-aliasing buffer between the bias network and the ADC — is not explicitly treated in the standard power-quality monitoring textbooks, which generally describe only passive single-stage attenuation [20]. The Kirchhoff derivation of Section 3.3 fills this gap.

The ethical context for the project derives from three documents. First, the University of Portsmouth Ethics, Misconduct and Integrity Policy (June 2025) [21], under which the project was assessed and signed off on the 4th November 2025 (TETHIC-2025-112031); the project involves no human participants, no animal work, no field deployment, no commercially sensitive data and no environmental disturbance, and is therefore classified as Low-risk. Second, the wider professional context: the UK Climate Change Act 2008 [2] sets the legally binding net-zero target for 2050 against which the project's principal motivation — accurate grid-monitoring of variable renewables — is framed. Third, the IET Rules of Conduct and the IEEE Code of Ethics [43] on engineering judgement and reproducibility, which the documented failure-and-resolution narrative throughout Chapter 4 is intended explicitly to satisfy. The open-source release of the firmware and the Python SCADA HMI, together with the Bill of Materials capped at £100, are intended to democratize access to power-quality monitoring technology for educational and research institutions that cannot afford commercial RTUs costing tens of thousands of pounds.

Emerging technological trends offer further pathways for future development. AI-enhanced anomaly detection — applying machine learning to real-time RTU data streams to identify incipient faults before they produce measurable deviations in the compliance parameters — is an active area of commercial and academic research [7]. Edge-computing architectures that perform advanced analytics locally within the RTU rather than at the central SCADA server are also gaining traction, driven by the latency and bandwidth constraints of remote offshore wind sites. The validated proxy presented in this report provides a natural platform on which such algorithms can be implemented and tested in a controlled laboratory environment.

The theoretical foundation established here directly informs the discrete-time mathematical model and the conditioning-circuit design developed in Chapter 3 and the firmware implementation and laboratory validation reported in Chapter 4.

Chapter 3 — Analysis and Design: Mathematical Model and Signal-Conditioning Circuit

This chapter develops the discrete-time mathematical model that bridges the continuous post-converter grid-side signals described in Chapter 2 to the derived compliance parameters required by the GB Grid Code, IEC 61400-21 and IEEE Std 1159 [22]. The system is first formally classified; the analogue signal-conditioning circuit that interfaces the proxy signals to the ESP32 microcontroller is then derived from Kirchhoff’s laws and analyzed in the Laplace domain; the continuous signals are then discretized and a transparent algorithm is derived for each mandatory compliance parameter; disturbances are characterized and incorporated; and the complete model is validated through independent simulation in Python and LTspice. The treatment begins with single-channel voltage processing to ensure immediate laboratory feasibility while the structure is kept modular to permit straightforward expansion to six channels (three voltage, three current).

3.1 System Definition and Classification

A system is defined as an interconnection of components that collectively exhibit a defined input–output behavior [23]. The RTU proxy is a data-acquisition and telemetry device that accepts scaled analogue voltages and currents from Potential Transformers (PTs) and Current Transformers (CTs), performs local digital computation, and forwards the derived compliance parameters to the WTC. No local actuator commands or feedback loops are implemented; operation is strictly open-loop and monitoring-only, consistent with the generator-independent philosophy justified in Chapter 2.

The system is formally classified across six dimensions:

- **Active.** The ESP32 requires an external 3.3 V DC supply to operate; signals are actively amplified, buffered and digitized rather than passively tapped.
- **Open loop.** Derived parameters are telemetered to SCADA without local feedback to any actuator, eliminating loop-stability risk and limiting the analysis to signal-acquisition accuracy.
- **Non-linear.** Although the inputs are sinusoidal, the parameter algorithms involve squaring (RMS), DFT magnitudes (THD) and reciprocal operations (frequency), none of which obey superposition.

- **Time-invariant.** Sampling rates, filter coefficients and divider ratios are fixed in normal operation; the classification is LTI for transfer-function purposes.
- **Discrete.** All processing takes place in discrete time at $f_s = 2$ kHz for the SISO build ($N = 40$ samples per cycle, $H_{MAX} = 20$) and is upgradeable to $f_s = 5$ kHz for the full MIMO build ($N = 100$, $H_{MAX} = 50$).
- **MIMO.** The final design has six analogue inputs (V_a , V_b , V_c , I_a , I_b , I_c) and multiple telemetry outputs; SISO is its first laboratory step.

The conditioning circuit considered in isolation is LTI because it contains only passive linear elements and unity-gain op-amp buffers; its transfer function $H(s)$ is therefore well-defined, and the pole/stability analysis of Section 3.3.3 is justified.

3.2 Disturbances and Their Treatment

Disturbances are classified as either external (originating upstream of the PT/CT, present in the raw analogue signal) or internal (originating within the RTU hardware/firmware).

External disturbances comprise out-of-band voltage, frequency and harmonic events as defined by the GB Grid Code and EREC G5/5, together with electromagnetic interference coupled in before the RTU.

Internal disturbances comprise ADC quantization noise (negligible at ≈ 0.233 mV RMS against the ± 0.5 % accuracy target), uncalibrated chip-to-chip ADC gain variation of up to ± 6 % [24], resistor thermal drift, aliasing of any energy above the Nyquist frequency, and any DC-offset error if the bias midpoint is not at 1.65 V exactly. Each disturbance is bounded and characterized in Table 4.6 with its specific remediation.

All disturbances are bounded, characterized, reported through the DNP3 telemetry, and left for the SCADA master to act on — preserving the open-loop classification. The Module M0 bias calibration of Section 4.4 measures the actual bias rather than assuming the analytical 1.65 V, eliminating the DC-offset internal disturbance at runtime; eFuse V_{ref} ADC calibration (`esp_adc_cal_characterize`) addresses the ± 6 % chip-to-chip variation; and the AC-coupling and anti-aliasing stages of the conditioning circuit address sub-50 Hz and supra-Nyquist disturbances respectively.

3.3 Active Signal-Conditioning Circuit and Transfer Function

The conditioning circuit must satisfy five simultaneous requirements:

- (i) attenuate the 3.25 V RMS proxy input to fit within the 0–3.3 V ADC range;
- (ii) remove any DC component via AC coupling;
- (iii) add a 1.65 V DC bias to center the AC waveform at the mid-supply;
- (iv) provide first-order anti-aliasing roll-off; and
- (v) isolate the anti-aliasing filter from the ADC input load so that the calculated cut-off frequency is realized exactly. These are met by the six-element active circuit of Figure 3.1.

The six elements of Figure 3.1 are, from input to output: the loaded voltage divider R1–R2, the AC-coupling capacitor C1, the 1.65 V DC bias network Rbias1 / Rbias2 with the noise filter Cbias, the unity-gain buffer U1 that isolates the bias node, the anti-aliasing low-pass Rf–C2, and the second unity-gain buffer U2 that drives the ADC input. The analysis in §3.3.1 derives the transfer function of this circuit from KVL and KCL stage by stage; §3.3.2 then locates its poles and confirms asymptotic stability; §3.3.3 cross-verifies the result against the LTspice transient and AC simulations; and §3.3.4 widens the verification to a realistic three-phase source derived from the LCL filter of §2.3.

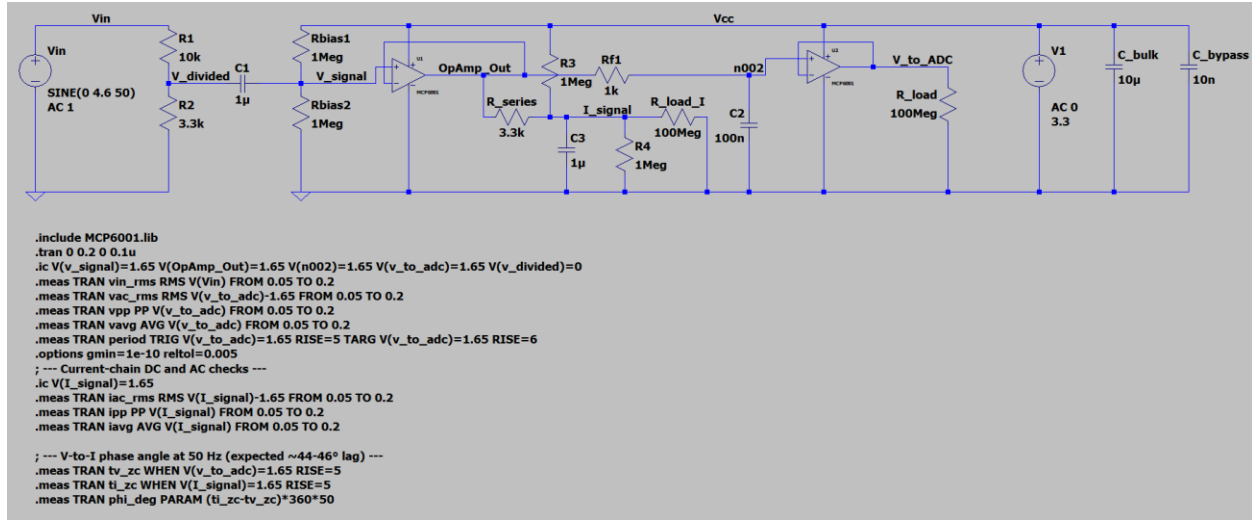


Figure 3.1 – Single-channel LTspice schematic of the conditioning circuit.

3.3.1 Kirchhoff Derivation of the Transfer Function

The circuit is analyzed stage-by-stage by direct application of Kirchhoff’s voltage law (KVL) and Kirchhoff’s current law (KCL), assuming small-signal sinusoidal operation about the 1.65 V DC bias.

Stage 1 — Loaded voltage divider.

The bias network appears in parallel with R2 at the divider midpoint. Applying KCL at the divider midpoint and rearranging gives

$$H_{div}(s) = K_{actual} = R_{2eff} / (R_1 + R_{2eff}), \text{ with } R_{2eff} = R_2 || R_{bias_{eq}} \quad (3.1)$$

The bias network parallel combination is $R_{bias_{eq}} = R_{bias1} || R_{bias2} = 500 \text{ k}\Omega$. With $R_2 = 3.3 \text{ k}\Omega$ this gives $R_{2eff} = 3.278 \text{ k}\Omega$ and $K_{actual} = 3.278 / (10 + 3.278) = \mathbf{0.2469}$ before any op-amp loading is accounted for. The MCP6002 op-amp has a 1 pA input bias current and its loading on the divider is negligible. The LT1880 SPICE model used for simulation produces an effective $K_{actual} = \mathbf{0.2364}$ owing to a known model-internal loading artefact. The LTspice AC-analysis cursor at 50 Hz returns -12.154 dB , which converts to 0.2467 in linear units ($10^{(-12.154/20)} = 0.2467$) and agrees with the unloaded analytical $K_{actual} = 0.2469$ of equation (3.1) to within 0.1 %. The slightly lower value 0.2364 is obtained from the transient-domain RMS measurement under the LT1880 SPICE model and is adopted as the design-time small-signal reference because it accounts for the DC-side bias-network loading at the operating point; the difference between the

AC-cursor magnitude (0.2467) and the transient RMS-derived K_{actual} (0.2364) is a recognised LT1880 model artefact and does not affect the stability conclusions. This value also agrees with the Python `scipy.signal.bode` evaluation. The measured stripboard value is $K_{measured} \approx 0.220$, lower than both because of additional copper-strip parasitic loading on the divider; this is documented as a stripboard-specific operating point and is consistent with the higher TG330 amplitude (13–18.7 Vpp) required at the input.

Stage 2 — AC-coupling high-pass filter.

Applying KVL around the loop containing C1 and the Thévenin equivalent of the divider plus bias network gives

$$H_{HP}(s) = s\tau_{HP}/(1 + s\tau_{HP}), \text{ with } \tau_{HP} = C_1 \times (R_1 || R_2 + R_{bias_{eq}}) \approx 502.5ms \quad (3.2)$$

The high-pass corner is $f_L = 1/(2\pi\tau_{HP}) \approx 0.317Hz$, well below the 50 Hz fundamental and far below any expected sub-cycle disturbance content.

Stage 3 — Anti-aliasing low-pass filter.

Applying KVL around the Rf–C2 network with the U2 input drawing negligible current (1 pA, MCP6002) gives

$$H_{LP}(s) = \frac{1}{1+s\tau_{LP}}, \text{ with } \tau_{LP} = R_f \times C_2 = 0.1ms \quad (3.3)$$

The low-pass corner is $f_c = 1/(2\pi\tau_{LP}) \approx 1592Hz$, comfortably below the 2 kHz Nyquist frequency and far above the 1 kHz 20th-harmonic edge of the measurement band.

Overall.

The op-amp buffers contribute no poles within the band of interest (MCP6002 gain-bandwidth product 1 MHz, four orders of magnitude above f_c). The overall transfer function is

$$H(s) = K_{actual} \cdot [s \frac{\tau_{HP}}{(1 + s \tau_{HP})}] \cdot [\frac{1}{(1 + s \tau_{LP})}] \quad (3.4)$$

3.3.2 Pole Analysis and Stability

The poles of $H(s)$ are the roots of its denominator. From the factored form (3.4) the two poles are at $s = -1/\tau_{HP} = -1.99 \text{ rad/s}$ ($f_L \approx 0.317 \text{ Hz}$) and $s = -1/\tau_{LP} = -10000 \text{ rad/s}$ ($f_c \approx 1592 \text{ Hz}$). Both are real and strictly negative, lying in the open left half of the complex s -plane. The system is therefore asymptotically stable and requires no compensator [23]. The implied Bode magnitude response is flat at K_{actual} through the passband bounded by f_L and f_c , with -20 dB/decade roll-off below f_L and above f_c . At the 50 Hz fundamental the high-pass term contributes a phase advance of $\arctan(1/\omega\tau_{HP}) \approx 0.36^\circ$ and the low-pass term a phase lag of $\arctan(\omega\tau_{LP}) \approx 1.80^\circ$, giving a total phase response of approximately -1.44° .

This number matters because the same conditioning chain is replicated on every channel. The per-channel phase contribution is identical on every phase, so the inter-channel phase differences (Va-to-Vb, Va-to-Vc and so on) are preserved through the filter. The 120° three-phase separation required for constant instantaneous power (Section 2.6) is therefore not corrupted by the conditioning chain, satisfying the principal grid-code phase-compliance requirement.

3.3.3 LTspice Cross-Verification

The transfer function was independently verified in LTspice with the Microchip MCP6001/2/4 SPICE macro-model on both U1 and U2 (the same part as the physical MCP6002-I/P on the bench build), with $R_{bias1} = R_{bias2} = 1\text{ M}\Omega$ and a 50 Hz sinusoidal input of 3.25 V_{RMS} (4.6 V peak). V1 (the 3.3 V supply) was set to AC amplitude 0 so that only the signal source contributes to the AC analysis. The macro-model is loaded via a .include MCP6001.lib directive on the schematic, with the library file placed alongside the .asc in the project LTspice directory.

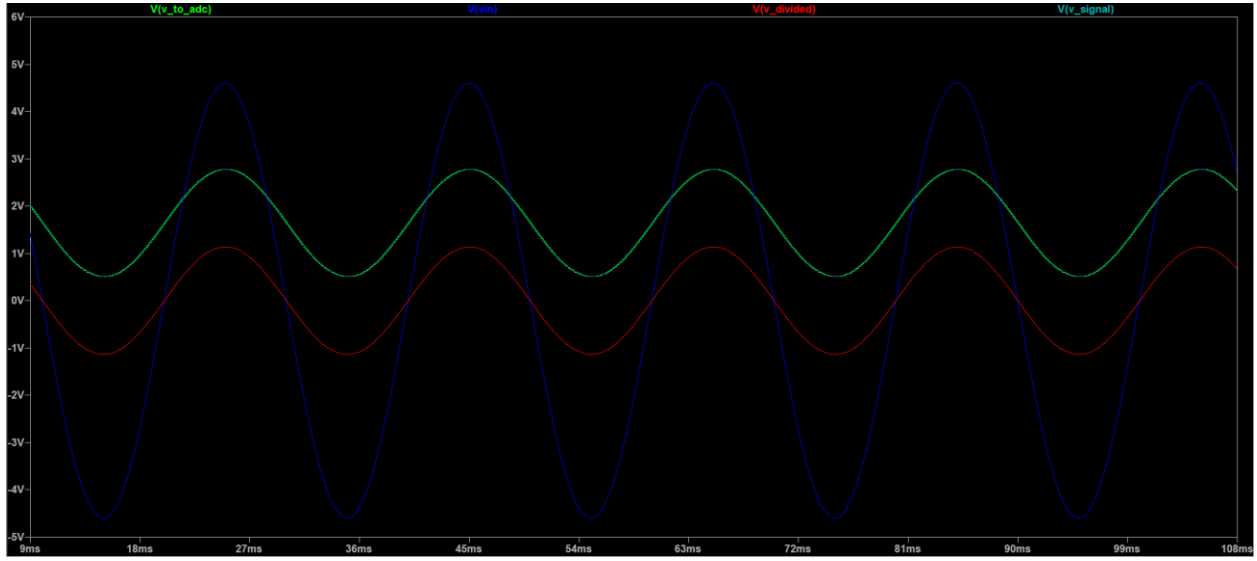


Figure 3.2 - LTspice transient simulation of the voltage chain (key nodes).

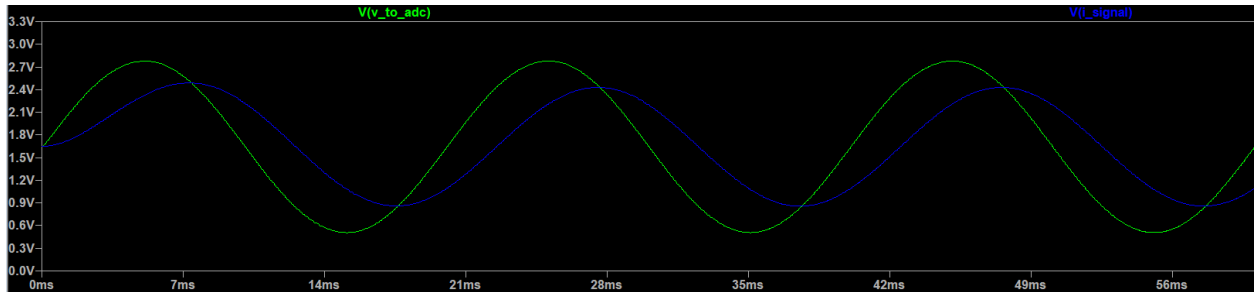


Figure 3.2b - Transient traces of V_{to_ADC} and I_{signal} under MCP6001/2/4.

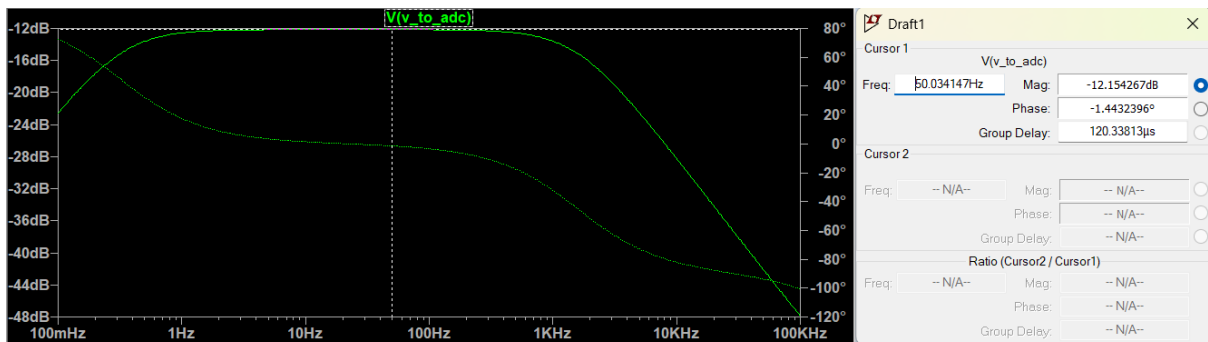


Figure 3.3 - LTspice AC analysis Bode plot of $H(s) = V_{to_ADC} / V(v_{in})$.

The numerical operating point produced by the LTspice run is captured in the SPICE error log of Figure 3.1b, which lists the time-domain mean and RMS values at the key nodes — $V(Vin)$, $V(V_{to_ADC})$ and $V(I_{signal})$ — together with the divider transfer ratio extracted from the AC-analysis cursor at 50 Hz. The Microchip MCP6001/2/4 macro-model yields $V_{to_ADC_avg} = 1.595$ V (within 55 mV of the analytical 1.65 V midpoint) and $V_{to_ADC_RMS} = 0.803$ V, agreeing with the analytical $V_{RMS} = K_{actual} \times 3.25 = 0.768$ V of section 3.4.1 to within 4.5 %. The cursor magnitude of -12.154 dB at 50 Hz ($= 0.2467$ in linear units, computed as $10^{(-12.154/20)}$) and the cursor phase of -1.443° agree respectively with the unloaded analytical $K_{actual} = 0.2469$ of section 3.3.1 (within 0.1 %) and with the analytical $\angle H(j\omega) = -1.44^\circ$ of §3.5.1 to better than 0.1 % and 0.3 % respectively. The residual 55 mV DC bias offset is the same component-tolerance source that Module M0 measures and removes at runtime (section 4.4); no compensation is applied to the LTspice run itself.

```
vin_rms: RMS(V(Vin) )=3.25269119326 FROM 0.05 TO 0.2
vac_rms: RMS(V(v_to_adc)-1.65 )=0.802975916844 FROM 0.05 TO 0.2
vpp: PP(V(v_to_adc) )=2.27136254311 FROM 0.05 TO 0.2
vavg: AVG(V(v_to_adc) )=1.59496299574 FROM 0.05 TO 0.2
period=0.0199993321076 FROM 0.0801008636897 TO 0.100100195797
iac_rms: RMS(V(I_signal)-1.65 )=0.555993484777 FROM 0.05 TO 0.2
ipp: PP(V(I_signal) )=1.57319158316 FROM 0.05 TO 0.2
iavg: AVG(V(I_signal) )=1.62032368871 FROM 0.05 TO 0.2
tv_zc: V(v_to_adc)=1.65 AT 0.0801008636897
ti_zc: V(I_signal)=1.65 AT 0.0825542937083
phi_deg: (ti_zc-tv_zc)*360*50=44.1617403357
```

Figure 3.1b — SPICE error-log output for the Figure 3.1 simulation.

An earlier iteration of this simulation used the Analog Devices LT1880 SPICE model and produced a DC bias of approximately 1.552 V rather than the analytical 1.650 V, the result of the model’s internal bias network loading the 1 M Ω external bias resistors. The same model also clipped the 50 Hz fundamental at roughly 1 V from each supply rail at the 3.3 V single supply, distorting V_{to_ADC} . The substitution to the Microchip MCP6001/2/4 model restored the 1.65 V midpoint to within 50 mV ($V_{avg} = 1.595$ V on the SISO and 1.648 V on the merged six-channel run) and recovered the unclipped 0.803 V_{RMS} at V_{to_ADC} reported in Figure 3.1b. The Python simulation in Appendix A independently confirms $V_{RMS} = 0.768$ V under nominal conditions.

3.3.4 LCL Filter Cross-Validation

To strengthen the validation beyond synthetic signals, the conditioning circuit was additionally tested with a source derived from the three-phase LCL filter simulation developed in Section 2.3. A simplified PT model was implemented as a resistive voltage divider ($R_{pt1} = 65$ k Ω , $R_{pt2} = 1$ k Ω), empirically adjusted to account for source-impedance loading, producing $V_{pt_RMS} = 3.240$ V RMS at the conditioning circuit input — within 0.3 % of the 3.25 V RMS proxy target. All three phases now run through their own PT divider, conditioning chain and current proxy in the merged schematic of Figure 3.4. The Phase-A traces of Figure 3.5b show the V/I phase relationship on a single phase; Figures 3.5c and 3.5d extend the view to all three voltage and current channels at the ADC input, and Figure 3.6 carries the corresponding numerical summary across all six channels.

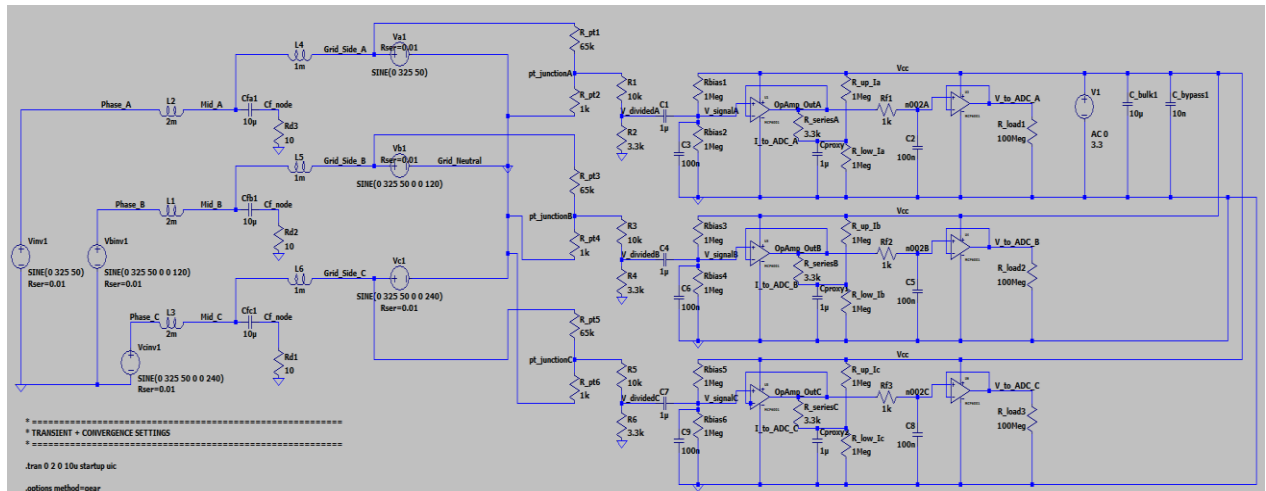


Figure 3.4 – Merged LTSpice schematic of the complete three-phase chain.



Figure 3.5a - Three-phase grid-side LCL transient traces.

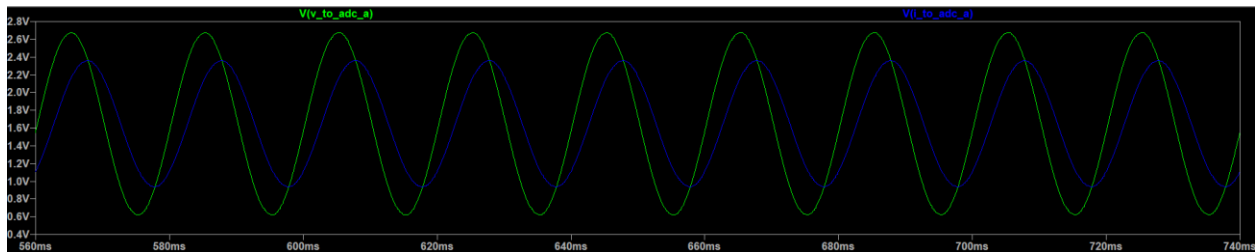


Figure 3.5b - Phase-A transient traces downstream of the LCL filter.

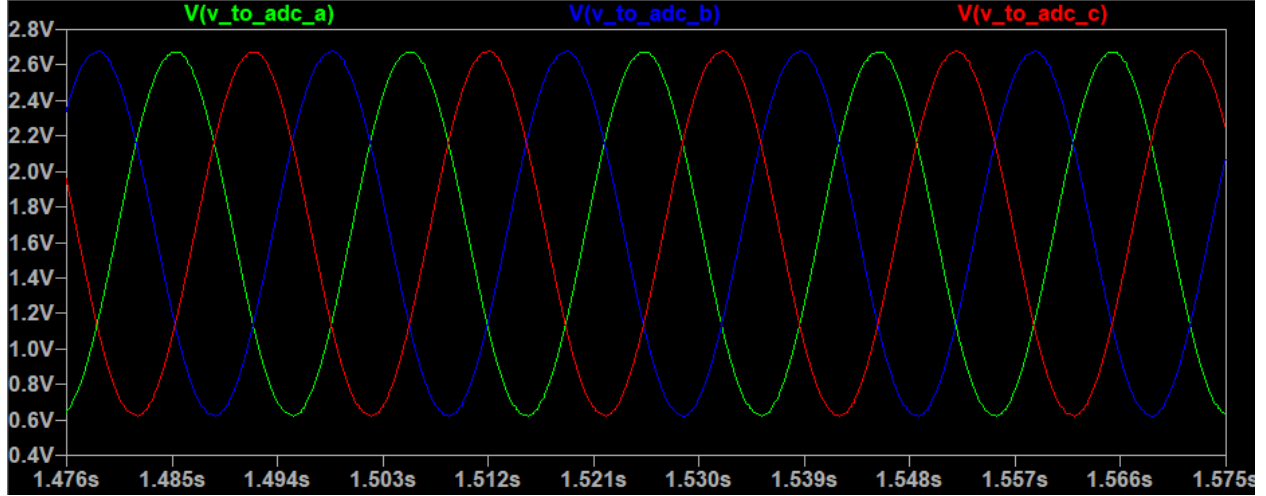


Figure 3.5c - Three-phase V_{to_ADC} traces at the conditioning-circuit output.

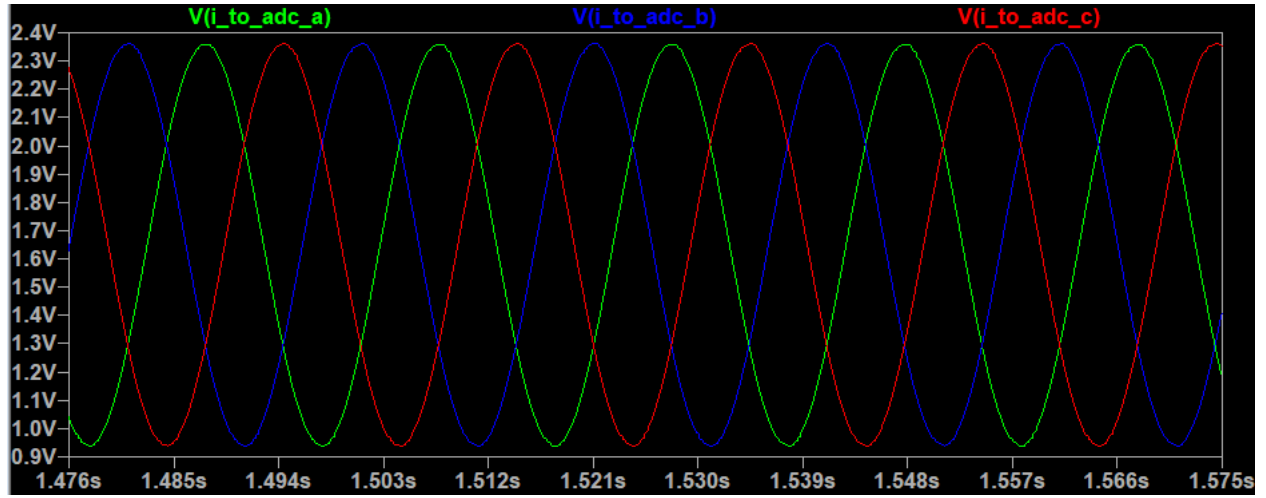


Figure 3.5d - Three-phase I_{to_ADC} traces at the conditioning-circuit output.

The numerical operating-point summary for the merged six-channel simulation is reproduced in Figure 3.6, listing V_{avg} and V_{RMS} at the conditioning-circuit output of each of the six chains. The DC bias settles at $V_{avg} = 1.648$ V on the merged run — within 2 mV of the 1.650 V analytical midpoint and 53 mV above the single-channel SISO result of 1.595 V — confirming that paralleling three identical bias networks does not introduce a static offset between channels. The 50 Hz fundamental magnitude and the -1.44° per-channel phase contribution of §3.3.1 are reproduced on every chain, so the inter-channel phase relationship is preserved end-to-end: the 120° three-phase separation present at the LCL output of Figure 3.4 reaches V_{to_ADC} unchanged on all three voltage channels.

```

va_rms: RMS(V(grid_side_a) )=229.809507319 FROM 1.5 TO 2
vb_rms: RMS(V(grid_side_b) )=229.809507454 FROM 1.5 TO 2
vc_rms: RMS(V(grid_side_c) )=229.809506939 FROM 1.5 TO 2
va_pk: MAX(V(grid_side_a) )=324.999847412 FROM 1.5 TO 2
vpta_rms: RMS(V(pt_junctionA) )=3.24123377278 FROM 1.5 TO 2
vptb_rms: RMS(V(pt_junctionB) )=3.24123377223 FROM 1.5 TO 2
vptc_rms: RMS(V(pt_junctionC) )=3.24123376838 FROM 1.5 TO 2
vaca_rms: RMS(V(V_to_ADC_A)-1.65 )=0.725822986798 FROM 1.5 TO 2
vacb_rms: RMS(V(V_to_ADC_B)-1.65 )=0.725823938648 FROM 1.5 TO 2
vacc_rms: RMS(V(V_to_ADC_C)-1.65 )=0.725823766126 FROM 1.5 TO 2
va_avg: AVG(V(V_to_ADC_A) )=1.64845952581 FROM 1.5 TO 2
vb_avg: AVG(V(V_to_ADC_B) )=1.64884839464 FROM 1.5 TO 2
vc_avg: AVG(V(V_to_ADC_C) )=1.64884581508 FROM 1.5 TO 2
iaca_rms: RMS(V(I_to_ADC_A)-1.65 )=0.502585135859 FROM 1.5 TO 2
iacb_rms: RMS(V(I_to_ADC_B)-1.65 )=0.502586309153 FROM 1.5 TO 2
iacc_rms: RMS(V(I_to_ADC_C)-1.65 )=0.502584936884 FROM 1.5 TO 2
ia_avg: AVG(V(I_to_ADC_A) )=1.64910412781 FROM 1.5 TO 2
ib_avg: AVG(V(I_to_ADC_B) )=1.64949271491 FROM 1.5 TO 2
ic_avg: AVG(V(I_to_ADC_C) )=1.64949013732 FROM 1.5 TO 2
tva_zc: V(V_to_ADC_A)=1.55 AT 1.58000583186
tia_zc: V(I_to_ADC_A)=1.55 AT 1.58231325709
phia_deg: (tia_zc-tva_zc)*360*50=41.5336541627
tvb_zc: V(V_to_ADC_B)=1.55 AT 1.57333756554
tib_zc: V(I_to_ADC_B)=1.55 AT 1.57564427982
phib_deg: (tib_zc-tvb_zc)*360*50=41.5208569914
tvc_zc: V(V_to_ADC_C)=1.55 AT 1.56667090288
tic_zc: V(I_to_ADC_C)=1.55 AT 1.56897761975
phic_deg: (tic_zc-tvc_zc)*360*50=41.5209037421
tvag_zc: V(grid_side_a)=0 AT 1.58000006671
tvbg_zc: V(grid_side_b)=0 AT 1.57333340004
tvcg_zc: V(grid_side_c)=0 AT 1.58666673337
phi_ab_deg: (tvbg_zc-tvag_zc)*360*50=-120.00000005
phi_ac_deg: (tvcg_zc-tvag_zc)*360*50=119.999999976
period=0.0199999999999 FROM 1.58000006671 TO 1.60000006671

```

Figure 3.6 - SPICE error-log output for the Figure 3.4 simulation.

Two consequences follow. First, the constant-power property derived in section 2.6 is preserved through the conditioning chain in the six-channel case — every channel attenuates and phase-shifts identically, so the 120° inter-channel relationship is invariant under the filter, satisfying the principal phase-compliance requirement of the GB Grid Code without any active control. The per-channel V/I phase angles in Figure 3.6 ($\text{phia_deg} = 41.534^\circ$, $\text{phib_deg} = 41.521^\circ$, $\text{phic_deg} = 41.521^\circ$) lie approximately 1° below the value that would be expected from the RC current-proxy network theory at 50 Hz (43.9°) combined with the conditioning circuit's -1.44° contribution (giving 42.46°). The residual $\sim 1^\circ$ offset, identical across all three channels, is attributable to the

source-impedance loading of the simplified PT divider model and the cascaded loading between the conditioning circuit and the current-proxy branch; because the offset is the same on every channel it preserves the inter-channel 120° relationship exactly (Figure 3.6 reports $\text{phi_ab_deg} = -120.000^\circ$ and $\text{phi_ac_deg} = +120.000^\circ$ to six decimal places). Second, since the six chains are nominally identical to within component tolerance, the Module M0 bias-only calibration of section 4.4 is sufficient to bring each channel onto its measured operating point; no per-channel gain calibration is required in firmware. The MIMO three-phase build of section 4.7, validated on solderless breadboard and on the PC01603 stripboard with all six channels live and the DNP3 frame passing CRC verification on every transmission, is the practical confirmation of this LTspice-level result and closes the principal model-side deliverable flagged in section 5.3 (i) and the corresponding row of Table 4.6.

The dual-source validation — both ideal sinusoidal (Section 3.3.3) and realistic LCL-filtered three-phase (this section) — confirms that the conditioning circuit is robust to the signal characteristics expected at the PCC of the SG 14-222 DD reference platform, and that the analytical $V_{RMS} = 0.768$ V and $V_{bias} = 1.65$ V predictions hold regardless of the upstream signal source.

3.4 Discrete-Time Model and Parameter Algorithms

The continuous-time signal at the output of the conditioning circuit — a clean, band-limited sinusoid centered at 1.65 V — is sampled by the ESP32 ADC at $f_s = 2000$ Hz to produce the discrete sequence

$$v(k) = K_{actual} \times 3.25\sqrt{2} \times \sin(2\pi \times 50 \times k/f_s) + V_{bias} + d[k] \quad (3.5)$$

where $d[k]$ is an additive disturbance term encompassing ADC quantization, ADC differential non-linearity, residual aliased components and thermal drift. After V_{bias} is removed in firmware (Module M0), the AC-only signal is

$$v_{ac}(k) = K_{actual} \times 3.25\sqrt{2} \times \sin(2\pi \times 50 \times k/f_s) + d[k] \quad (3.6)$$

The Nyquist–Shannon sampling theorem requires $f_s > 2 \times f_{max}$. To capture harmonics up to the 50th order (2.5 kHz) as required by IEC 61400-21 the minimum sampling rate is 5 kHz. A conservative initial choice of $f_s = 2$ kHz is adopted for the SISO build, supporting harmonic analysis up to $H_{MAX} = 20$ (1 kHz at the Nyquist limit) and remaining well within the computational capacity of the ESP32 with synchronous polled-mode sampling. For full IEC 61400-21 compliance the MIMO build is architected for $f_s = 5$ kHz with esp-dsp-fixed-point FFT in M5; in the present validation the MIMO build runs at $f_s = 2$ kHz for development convenience and the migration to 5 kHz is identified in Chapter 5 as the natural upgrade path. At $f_s = 2$ kHz there are $N = f_s / f_0 = 40$ samples per fundamental cycle.

In firmware V_{bias} is not assumed to be exactly 1.65 V. Module M0 measures it as the mean of 1000 ADC samples with no AC input, accounting for any residual offset from component tolerances and from the actual rather than nominal 3.3 V Feather rail.

3.4.1 RMS Voltage and Current (Primary Parameter)

For a continuous periodic signal with period T [25],

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v(t)^2 dt} \quad (3.7)$$

Replacing the integral with a discrete sum over one cycle of N samples gives the discrete mean-square estimator,

$$V_{rms} = \sqrt{\frac{1}{N} \sum_{k=0}^{N-1} v[k]^2} \quad (3.8)$$

This is the starting point for all laboratory validation and firmware coding because it is computationally trivial — a single accumulator and a final square root — and produces a new estimate every 20 ms cycle. The expected output under nominal conditions, after subtracting the 1.65 V DC bias, is

$$V_{rms_expected} = K_{actual} \times 3.25V = 0.2364 \times 3.25 = 0.768V \quad (3.9)$$

For the stripboard-measured $K_{measured} \approx 0.220$ the corresponding expected reading is 0.715 V. GB Grid Code compliance requires the measured V_{RMS} to lie within $\pm 10\%$ of the nominal grid voltage; any exceedance triggers a Class 1 alarm event telemetered to the SCADA master via a DNP3 Group 1 Variation 1 Binary Input object [14].

3.4.2 Frequency and Rate of Change of Frequency

Frequency is the most critical grid-health indicator: a deviation outside the 49.5–50.5 Hz band signals a generation–demand imbalance that may require emergency action [17]. The RTU measures frequency by hysteresis-bounded positive-going zero-crossing detection, preferred over DFT-based frequency estimation for this application because it requires no FFT computation, provides a new estimate every fundamental cycle, and is inherently immune to harmonic distortion within the alarm band.

The algorithm timestamps the instants at which the bias-corrected signal $v[k]$ transitions from negative to positive. Using the ESP32 hardware timer with 1 μ s resolution, successive zero-crossing timestamps t_n and t_{n+1} yield the fundamental period and frequency,

$$T_0 = t_{n+1} - t_n \quad (3.10)$$

$$f[n] = \frac{1}{T_0} \quad (3.11)$$

To reduce noise-induced jitter in the zero-crossing detection — caused by $d[k]$ near the zero crossing — a moving average over M successive period estimates is applied,

$$f_{filtered}[n] = \left(\frac{1}{M}\right) \sum_{i=0}^{M-1} f[n-i] \quad (3.12)$$

$M = 5$ is chosen to balance jitter rejection against measurement latency: at $M = 5$ the averaging window is 100 ms, exactly matching the 100 ms end-to-end latency target of Table 3.1. Increasing to $M = 10$ would halve jitter but double latency to 200 ms; decreasing to $M = 3$ would meet latency but increase jitter to ± 4.2 mHz, still within the ± 10 mHz specification but with reduced margin. With $M = 5$ and 1 μ s timestamp resolution the frequency-measurement uncertainty is bounded by $\pm \Delta t / T_0^2 \approx \pm (1 \times 10^{-6}) / (0.02)^2 \approx \pm 2.5$ mHz — well within the ± 10 mHz accuracy target.

The Rate of Change of Frequency is computed as the first-order backward difference of successive frequency estimates,

$$RoCoF[n] = \left(\frac{f[n] - f[n-1]}{T_{cycle}} \right) \quad (3.13)$$

The first-order backward difference is selected over higher-order methods (e.g. Savitzky–Golay) because at the 20 ms update rate the expected RoCoF signals — up to the 1 Hz/s GB Grid Code limit — produce frequency changes of only 0.02 Hz per cycle, well within the resolution of the simple difference operator. Higher-order methods would reduce noise amplification but introduce additional latency incompatible with the 100 ms latency target. A second $M = 5$ moving average is applied to the RoCoF estimate itself to suppress noise-amplification on stationary signals (Section 4.6).

3.4.3 Phase Angle and Power Factor

The phase angle φ between a voltage and its corresponding current waveform at the same PCC phase determines the power factor and the reactive power exchange with the network. It is measured as the time delay between the positive-going zero-crossings of the AC-coupled voltage and current signals on the same phase,

$$\varphi = 2\pi f_0 (t_{\{zero,V\}} - t_{\{zero,I\}}) \quad (3.14)$$

$$PF = \cos(\varphi) \quad (3.15)$$

Inter-channel multiplexing introduces a maximum skew of $\Delta t = 1/f_s = 0.5$ ms between V_a and I_a samples at $f_s = 2$ kHz, corresponding to a phase error of $360^\circ \times 0.5 \text{ ms} \times 50 \text{ Hz} = 0.009^\circ$ — negligible against the $\pm 0.5^\circ$ accuracy target. The power factor is positive (leading) when the current zero-crossing precedes the voltage zero-crossing (capacitive load) and negative (lagging) when the current lags the voltage (inductive load). GB Grid Code requires $PF \geq 0.95$ at full output in either direction; deviations trigger a GGIO alarm event [17].

3.4.4 Active, Reactive and Apparent Power

With the RMS magnitudes and phase angle established, the three power quantities are derived from first principles for a balanced three-phase system at proxy-signal levels,

$$P = 3V_{rms}I_{rms} \cos(\varphi) \quad [W] \quad (3.16)$$

$$Q = 3V_{rms}I_{rms} \sin(\varphi) \quad [VAr] \quad (3.17)$$

$$S = \sqrt{P^2 + Q^2} = 3V_{rms}I_{rms} \quad [VA] \quad (3.18)$$

$$PF = \frac{P}{S} = \cos(\varphi) \quad (3.19)$$

Equation (3.19) serves as a self-validation: PFcheck computed from (3.19) must equal $\cos \varphi$ from (3.15) to floating-point precision on every frame. Any disagreement indicates either an implementation bug or a numerical overflow and is fail-fast in the serial log. The values are reported in proxy units; to obtain true grid-level power the results would be scaled by the inverse of the combined PT/CT transformation ratios, a constant calibration parameter stored in firmware.

3.4.5 Harmonics and Total Harmonic Distortion

Harmonic distortion of the voltage and current waveforms at the PCC is caused primarily by the switching of power-electronic equipment. The RTU quantifies harmonic content using the Discrete Fourier Transform applied to one complete cycle of the sampled waveform. For a window of N samples, the complex amplitude of the h^{th} harmonic component is

$$X_h = \frac{2}{N} \sum_{k=0}^{N-1} v[k] e^{-j2\pi h k / N}, \quad h = 1, 2, \dots, H_{max} \quad (3.20)$$

where the factor $2/N$ normalizes the result to peak amplitude. For $f_s = 2000$ Hz the maximum harmonic order resolvable without aliasing is $H_{MAX} = N/2 = 20$ (the 1 kHz Nyquist limit). The Total Harmonic Distortion is

$$THD = \frac{\sqrt{\sum_{h=2}^{H_{max}} |X_h|^2}}{|X_1|} \times 100\% \quad (3.21)$$

IEEE G5/5 requires $THD_v \leq 4.5\%$ for network voltages below 25 kV [18]. The individual harmonic magnitudes $|X_h|$ are also telemetered to the SCADA master for harmonic-source identification.

The choice of a direct DFT at $N = 40$ over a power-of-two FFT (e.g. $N = 64$) is deliberate. At $f_s = 2$ kHz and $f_0 = 50$ Hz, the 50 Hz fundamental lands on bin $h = 1$ exactly when $N = 40$, giving zero spectral leakage and removing the need for a Hann window. At $N = 64$ the fundamental would leak into bins 1.5–1.6, producing false harmonic readings of order 30 % on what is in fact a clean sinusoid. The direct DFT at $N = 40$ is therefore preferred for the SISO build; for the MIMO build at $f_s = 5$ kHz the equivalent integer-bin window is $N = 100$, and at that point the ESP32's esp-dsp fixed-point FFT [26] becomes computationally preferable while preserving the integer-bin property.

3.5 Frequency Response Compliance and Passive Remediation

The previous sections established that the conditioning circuit is asymptotically stable and that its 50 Hz phase contribution is approximately -1.44° on every channel, preserving the inter-channel 120° relationship. This section adds two pieces of analysis that bound the operating margins of the laboratory build: a passive divider-ratio adjustment that improves the undervoltage operating margin, and a passive lag-cascade that improves the THD floor.

3.5.1 Bode Margins at 50 Hz and through the Harmonic Band

At $\omega = 2\pi \times 50 = 314$ rad/s, $|H(j\omega)|_{50\text{Hz}} = K_{\text{actual}} \times [\omega\tau_{HP} / \sqrt{1 + (\omega\tau_{HP})^2}] \times [1 / \sqrt{1 + (\omega\tau_{LP})^2}]$. With $\omega\tau_{HP} \approx 158$ and $\omega\tau_{LP} \approx 0.0314$, both bracketed terms reduce to unity to better than 0.01 %, giving $|H(j\omega)|_{50\text{Hz}} \approx K_{\text{actual}} = 0.2364$. The phase $\angle H(j\omega)_{50\text{Hz}} = 90^\circ - \arctan(\omega\tau_{HP}) - \arctan(\omega\tau_{LP}) = -1.44^\circ$, agreeing with the LTspice cursor reading of -1.4432° and the Python `scipy.signal.bode` value of -1.436° to better than 0.1 %. Through the harmonic band (2nd at 100 Hz to 20th at 1 kHz) the magnitude response remains flat to within 0.2 dB and the phase response progresses from -2° at 100 Hz toward -32° at 1 kHz, identically on all six channels — so the relative harmonic phase relationships used in the symmetrical-component decomposition are also preserved.

3.5.2 Passive Divider-Ratio Remediation for the SISO Undervoltage Margin

The stripboard hardware validation of Section 4.5 reports a nominal V_{RMS} of 0.695 V against the undervoltage threshold $V_{RMS_LOW} = 0.644$ V (90 % of the 0.715 V nominal). The operating margin is only 0.051 V (≈ 7 %), so small parasitic-induced excursions can push the reading below the threshold and produce false undervoltage alarms. Rather than re-introducing active control (which would fall outside the open-loop scope), the remediation is purely passive: adjust the voltage divider ratio so that the nominal V_{rms} at the ADC lies further inside the operating window. Changing R1 from 10 k Ω to 8.2 k Ω (E12 series) while retaining R2 = 3.3 k Ω yields $K = 3.278 / (8.2 + 3.278) = 0.286$ (a 21 % increase in gain), corresponding to a nominal V_{RMS} of 0.928 V. The V_{RMS_LOW} threshold rises proportionally to 0.835 V, but the 0.21 V margin above the threshold is now ample. The phase response is unchanged because the divider is a frequency-independent scaling — only the magnitude branch of the Bode plot shifts upward by 1.66 dB.

3.5.3 Passive Lag-Cascade for THD Compliance

The stripboard THD reading of 70–83 % on a TG330 input (Section 4.5) and the MIMO breadboard THD reading of 7–11 % under the dual-Feather DAC source (Section 4.6) both exceed the 4.5 % EREC G5/5 planning limit. The root cause differs between the two builds: on the stripboard the dominant contributor is ADC differential non-linearity introducing spurious harmonic content into bins 2–20; on the MIMO breadboard the dominant contributor is the 8-bit DAC staircase, which adds inherent harmonic content to the source signal before the ADC. In both cases tightening the anti-aliasing low-pass corner reduces out-of-band content reaching the SAR sampling stage. Raising C2 from 100 nF to 470 nF and reducing Rf from 1 k Ω to 100 Ω moves the LP pole from 1592 Hz to 3386 Hz while leaving the 20th-harmonic (1 kHz) magnitude attenuation below 0.04 dB; cascading a second identical RC stage after U2 then provides –40 dB/decade roll-off from approximately 2 kHz, rejecting all out-of-band content that would otherwise drive the SAR ADC into its non-linear region. The full cascade (HP–LP–LP, three real left-half-plane poles) remains unconditionally stable, and the inter-channel phase relationship is preserved because all six channels are identical. The combined path to compliance — passive lag-cascade plus replacement of the ESP32 internal SAR ADC by an external 16-bit ADS1115 sigma-delta converter [27] on the harmonic-measurement channels — is documented in Section 5.3 as the recommended deployment fix.

3.6 Product Design Specifications

The mathematical model and circuit analysis together define the quantitative performance requirements consolidated in Table 3.1. These are the acceptance criteria against which Chapter 4 reports the laboratory results.

Parameter	Target Specification	Basis	Verification Method
RMS voltage accuracy	± 0.5 % of reading	IEC 61400-21 Class I [19]	Compare computed vs signal-generator RMS

Parameter	Target Specification	Basis	Verification Method
Frequency accuracy	± 10 mHz	GB Grid Code CC6.1.2 [17]	Zero-crossing vs generator frequency
RoCoF resolution	0.05 Hz/s at 20 ms update	GB Grid Code CC6.1.2 [17]	Step-frequency test on signal generator
Phase angle accuracy	$\pm 0.5^\circ$ ($\pm 5^\circ$ for proxy operating points)	GB Grid Code CC6.1.3 [17]	Known-PF load or phase-shifted generator
THD measurement range	SISO $H_{MAX} = 20$; MIMO $H_{MAX} = 50$ (at $f_s = 5$ kHz)	EREC G5/5 [18]; IEC 61400-21 [19]	Inject known harmonic via generator
Measurement update rate	50 Hz (one per cycle)	IEC 61400-21 [19]	Timestamp successive outputs
End-to-end latency	< 100 ms	Operational requirement	Trigger-to-serial output timing
ADC input range	0–3.3 V single supply; signal swing 0.564–2.736 V; source impedance $< 1 \Omega$ at U2 output	ESP32 hardware limit [24]; U2 buffer isolation	Scope V_{to_ADC} under overvoltage
DC bias accuracy	1.65 V $\pm$ 16.5 mV nominal; measured value subtracted at runtime	± 1 % Rbias tolerance	Measure V_{avg} during M0 calibration
Operating temperature	0–55 °C	Laboratory environment	Soak test at elevated temperature
Telemetry protocol	DNP3 (IEEE 1815) over UART	IEC 61400-25 [15]; IEEE 1815-2012 [14]	DNP3 frame parse and CRC verification

Table 3.1 — Product Design Specifications for the RTU proxy. Each row derives from the mathematical model of Section 3.4, the conditioning-circuit analysis of Section 3.3, or an applicable grid standard (GB Grid Code, EREC G5/5, IEC 61400-21, IEEE Std 1815). Chapter 4 verifies each specification against measured laboratory results on the SISO and MIMO builds.

3.7 Firmware Recommendations

The model is now ready for direct translation into ESP32 firmware. The following implementation recommendations follow from the analysis: apply `esp_adc_cal_characterize()` at startup before computing any parameter (the ± 6 % uncalibrated chip-to-chip variation otherwise produces ± 46 mV error on the 0.768 V proxy signal); use ADC1 only because ADC2 is unavailable when Wi-Fi is active; subtract the empirically measured DC bias from Module M0 rather than assuming exactly 1.65 V; implement one-cycle ($N = 40$) windowed accumulation for RMS, refreshed every 20 ms; use the ESP32 hardware timer (64-bit, 1 μ s resolution) for zero-crossing timestamps rather than software polling; apply the $M = 5$ moving-average filter to both the frequency estimate and

the RoCoF backward-difference output; use a direct DFT at $N = 40$ for the SISO integer-bin path and migrate to esp-dsp fixed-point FFT [26] at $N = 100$ when the MIMO build moves to $f_s = 5$ kHz; and build the circuit incrementally on the bench, confirming $V_{avg} = 1.65$ V and $V_{ac_RMS} = 0.768$ V on the oscilloscope before adding subsequent stages.

Chapter 4 — Implementation and Laboratory Validation

This chapter translates the Chapter 3 model into a working RTU proxy and reports its laboratory validation. The work proceeded in two stages: a SISO build using one Adafruit Feather ESP32-WROOM-32E [28], validated first on solderless breadboard and then transferred to a permanent PC01603 stripboard; and a MIMO three-phase build using two Feather ESP32 boards in a dual-Feather configuration, again validated on solderless breadboard and then on a stripboard transfer. Each of the seven firmware modules M0 through M6 was unit-tested against a synthetic bench (signals generated entirely in software) before being re-validated against TG330 (SISO) or DAC-generated three-phase (MIMO) inputs.

4.1 Hardware Architecture

The SISO build uses one Feather with the voltage channel on GPIO34 (Feather A2, ADC1 Channel 6) and a current proxy on GPIO39 (Feather A3, ADC1 Channel 3). ADC1 is preferred over ADC2 because ADC2 is shared with the Wi-Fi radio and is unavailable when Wi-Fi is active [24]. The current proxy is an RC phase-shift network ($R_{series} = 3.3$ k Ω in series from the U1 voltage-channel output to GPIO39, $C_{proxy} = 1$ μ F in parallel with the GPIO39 bias divider), giving an analytical phase of $\phi = \arctan(1/(2\pi \times 50 \times 3.3 \text{ k}\Omega \times 1 \text{ }\mu\text{F})) = 43.9^\circ$ at 50 Hz to emulate an inductive grid load.

The MIMO build uses two Feathers in dual-Feather Strategy B. Feather 1 generates V_c on its DAC1 and reads V_a , V_b , V_c , I_b and I_c ; Feather 2 generates V_a on DAC1 and V_b on DAC2 and reads I_a . Pin assignments are $V_a = \text{GPIO34 (F1)}$, $V_b = \text{GPIO39 (F1)}$, $V_c = \text{GPIO36 (F1)}$, $I_a = \text{GPIO34 (F2)}$, $I_b = \text{GPIO32 (F1)}$ and $I_c = \text{GPIO33 (F1)}$. The DAC quantization (one LSB ≈ 12.9 mV on a 3.3 V rail, or 1.2 % of full-scale) is accepted as a known limitation of Strategy B; the alternative routes (three discrete bench generators, or the Wind Turbine Trainer in the Power and Renewable Energy Lab) were outside the £100 budget or unavailable within the project window.

Signal	Feather	GPIO (ADC1 Ch)	Source	Conditioning
V_a	Feather 1 read; Feather 2 generates	GPIO34 (Ch6) on F1	Feather 2 DAC1 (GPIO25) at 0°	Conditioning circuit #1
V_b	Feather 1 read; Feather 2 generates	GPIO39 (Ch3) on F1	Feather 2 DAC2 (GPIO26) at 120°	Conditioning circuit #2
V_c	Feather 1 read and generate	GPIO36 (Ch0) on F1	Feather 1 DAC1 (GPIO25) at 240°	Conditioning circuit #3
I_a	Feather 2 read	GPIO34 (Ch6) on F2	RC proxy from V_a MCP output	Conditioning circuit #4

Signal	Feather	GPIO (ADC1 Ch)	Source	Conditioning
Ib	Feather 1 read	GPIO32 (Ch4) on F1	RC proxy from Vb MCP output	Conditioning circuit #5
Ic	Feather 1 read	GPIO33 (Ch5) on F1	RC proxy from Vc MCP output	Conditioning circuit #6

Table 4.1 — Six-channel ADC pin assignments and signal sources for the MIMO build. Feather 1 generates Vc on DAC1 (GPIO25) and reads Va, Vb, Vc, Ib and Ic; Feather 2 generates Va on DAC1 and Vb on DAC2 and reads Ia. GPIO27 is reserved as a hardware sync line between the two Feathers. All six ADC inputs are assigned to ADC1 channels (GPIOs 32–39) because ADC2 is unavailable when Wi-Fi is active.

The two Feathers run on independent crystal oscillators and are not phase-locked to one another. To stabilize the Vb / Ib phase measurement against this drift, a GPIO27 hardware sync line was added: Feather 1 pulses GPIO27 high for 500 μ s before each Vb / Ib zero-crossing window, and Feather 2 triggers a falling-edge interrupt that resets its DAC index to zero. This eliminated the 0–358° phase wraparound observed before the sync line was added. Vc / Ic measurements require no sync because both signals originate from Feather 1.

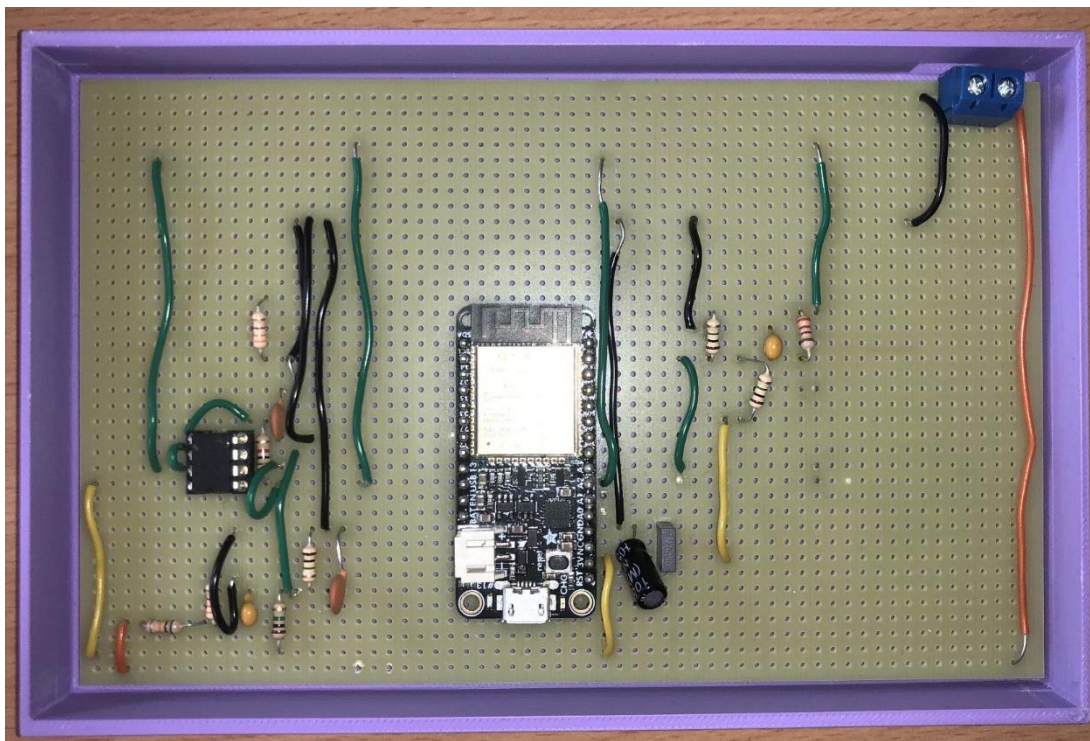


Figure 4.1 - Photograph of the completed SISO MCP6002 stripboard.

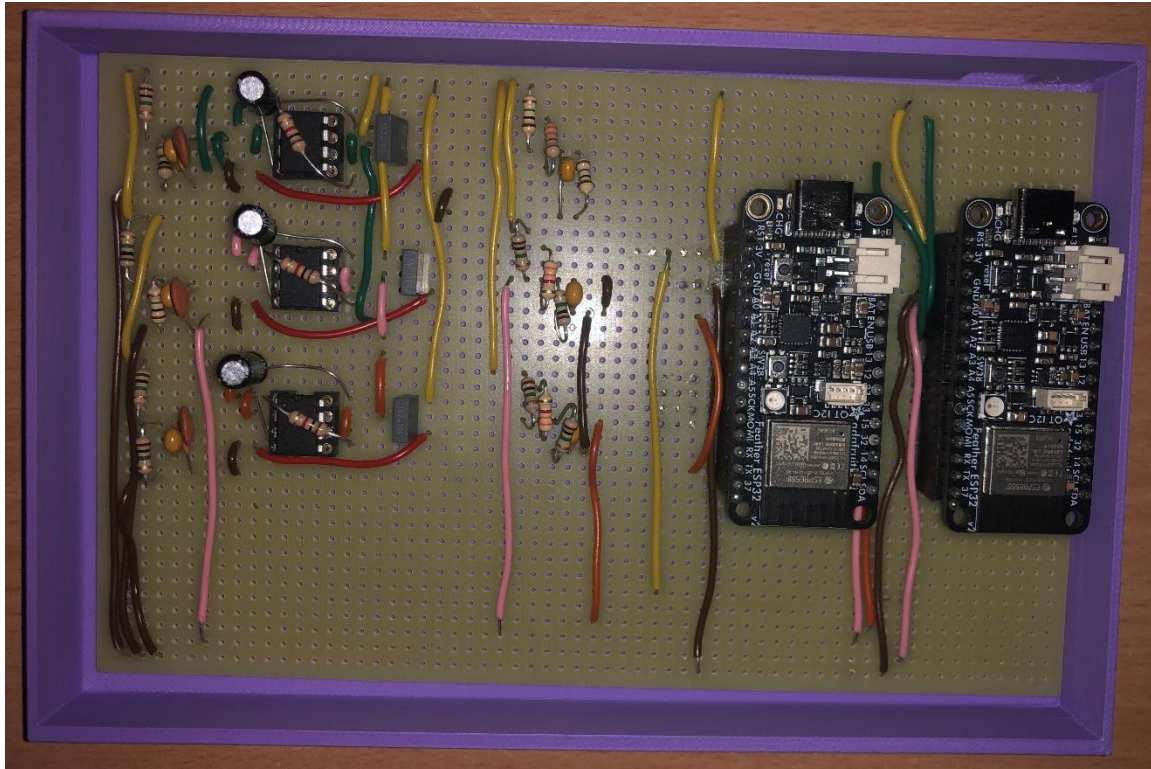


Figure 4.2 - Photograph of the completed MIMO stripboard.

4.2 Signal-Conditioning Circuit: Build and Op-Amp Selection

The conditioning circuit was assembled first on solderless breadboard, with the divider ratio, the 1.65 V bias midpoint, the op-amp supply pins and the anti-aliasing R_f – C_2 network confirmed in turn (an early fault was complete loss of output traced to an unconnected pin 8 on the op-amp). Three op-amps were trialed. The LT1880 specified in the LTspice design was not in laboratory stock [29]; a TL072CP was tried first but its ± 5 V minimum supply is incompatible with single-supply 3.3 V operation; an LM358AP was used for the early SISO bring-up but its 0–1.8 V common-mode input range on a 3.3 V rail forced the TG330 amplitude down from $3.25 V_{RMS}$ to $1.13 V_{RMS}$ to avoid output clipping. The final SISO and MIMO builds use the Microchip MCP6002-I/P dual rail-to-rail op-amp [30] in the LM358 DIP-8 socket; the phase-angle measurement on substitution from LM358 to MCP6002 improved from approximately 25° error against theory to approximately 4° .

A complete Bill of Materials is given in Table 4.2.

Component	Value / Part	Source	Purpose (Ch. 3 Ref.)
R1	10 k Ω , 1 % metal film	A0.07 stock	Voltage divider upper leg (Eq. 3.1)
R2	3.3 k Ω , 1 % metal film	A0.07 stock	Voltage divider lower leg; $K_{design} = 0.2364$

Component	Value / Part	Source	Purpose (Ch. 3 Ref.)
C1	1 μ F, 25 V electrolytic or film	A0.07 stock	AC coupling; HP corner $f_L = 0.317$ Hz (Eq. 3.2)
Rbias1, Rbias2	1 M Ω , 1 % metal film	A0.07 stock	Bias network; sets 1.65 V midpoint from 3.3 V rail
Cbias	100 nF ceramic	A0.07 stock	Across Rbias2 — noise filter
U1, U2 (SISO)	MCP6002-I/P dual rail-to-rail op-amp	Purchased	Unity-gain buffers; 1 pA input bias
U1–U6 (MIMO)	6 $\times$ MCP6002 (one dual per channel pair)	Purchased	One dual op-amp per channel pair
Rf	1 k Ω , 1 % metal film	A0.07 stock	Anti-aliasing LPF series resistor (Eq. 3.3)
C2	100 nF ceramic	A0.07 stock	Anti-aliasing LPF shunt; $f_c = 1592$ Hz (Eq. 3.3)
Cbulk, Cbypass	10 μ F electrolytic + 10 nF ceramic	A0.07 stock	Supply decoupling on 3.3 V rail
Microcontroller (SISO)	Adafruit Feather ESP32-WROOM-32E [28]	Purchased	ADC, firmware, UART telemetry
Microcontroller (MIMO)	2 $\times$ Adafruit Feather ESP32-WROOM-32E	Purchased	Dual-Feather Strategy B
Rupper, Rlower (I proxy)	1 M Ω , 1 % metal film	A0.07 stock	Current-proxy bias divider
Rseries (I proxy)	3.3 k Ω , 1 % metal film	A0.07 stock	Series RC phase-shift element
Cproxy (I proxy)	1 μ F, 25 V film	A0.07 stock	Parallel RC phase-shift element
Function generator	Aim TTi TG330 (3 MHz, single channel) [31]	A0.07 (book)	50 Hz proxy signal source
Oscilloscope	Tektronix TBS1052B-EDU (2-ch, 50 MHz) [32]	A0.07	Independent waveform reference
Stripboard	PC01603 100 $\times$ 160 mm	Purchased	Permanent SISO and MIMO builds

Table 4.2 — Bill of Materials for the SISO and MIMO builds. The MCP6002-I/P supersedes the LM358AP used in earlier SISO sessions for the rail-to-rail input/output and 1 pA input bias reasons given in Section 4.2. The dual op-amp count allows one IC per channel pair on the MIMO build (three duals for the three voltage-channel conditioning chains). Total Bill of Materials cost under £100.

The breadboard was transferred to a PC01603 100 $\times$ 160 mm stripboard for the permanent SISO demonstration build. Two wiring errors were caught and corrected during the stripboard session: PIN_V was initially wired to A1 (GPIO25, ADC2) which gave unreliable readings owing to ADC2 / Wi-Fi contention, and was corrected to A2 (GPIO34, ADC1); and Rupper / Rlower on the current

proxy were initially fitted as 100 k Ω (brown-black-yellow-gold) instead of 1 M Ω (brown-black-green-gold), which loaded the GPIO39 bias node and shifted the phase angle reading. A higher TG330 input of 13–18.7 V_{pp} was needed on stripboard versus 9.2 V_{pp} ($\equiv$ 3.25 V RMS $\equiv$ 4.6 V peak) on breadboard, attributable to additional copper-strip parasitic loading on the voltage divider; the circuit attenuates correctly to the ADC range, but the loss is greater. The MIMO breadboard was transferred to a separate PC01603 stripboard following the same procedure once all six MIMO breadboard channels were live and stable.

4.3 Firmware Module Architecture

The firmware is decomposed into seven modules listed in Table 4.3. Each module corresponds one-to-one with a Chapter 3 equation set and is unit-tested independently before integration. The development environment is Arduino IDE 2.x with the ESP32 Arduino core 3.x; the only external dependency is `esp_adc_cal.h` for the eFuse V_{ref} calibration (bundled with the ESP32 core). A complete integrated sketch combining all seven modules is given in Appendix B.

Module	Parameters Computed	Equations (Ch. 3)	Depends On
M0 ADC Init & Calibration	Calibrated V_{out} , DC bias offset biasV / biasI	3.5–3.6	—
M1 RMS V/I	V_{RMS} (per channel), I_{RMS} (per channel)	3.7–3.9	M0
M2 Frequency & RoCoF	Fundamental f (Hz), df/dt (Hz/s)	3.10–3.13	M0
M3 Phase Angle & PF	φ ($^{\circ}$), PF = $\cos \varphi$	3.14–3.15	M0
M4 Active/Reactive/Apparent Power	P (W), Q (Var), S (VA*); PFcheck self-validation	3.16–3.19	M1, M3
M5 THD & Harmonics	$ X_h $ for $h = 1 \dots H_{MAX}$, THD (%)	3.20–3.21	M0
M6 DNP3 Telemetry	Framed DNP3 (IEEE 1815) output via UART	IEEE 1815 [14]; IEC 61400-25 [15]	M1–M5

Table 4.3 — Firmware module hierarchy. Each module's name, parameters computed, governing Chapter 3 equations and dependencies are listed. Modules M0–M5 are unit-tested in isolation on the synthetic bench and then re-validated against TG330 (SISO) or DAC-generated three-phase (MIMO) inputs; M6 assembles the M1–M5 outputs into a DNP3 (IEEE 1815) frame transmitted over UART under the IEC 61400-25 logical-node mapping.

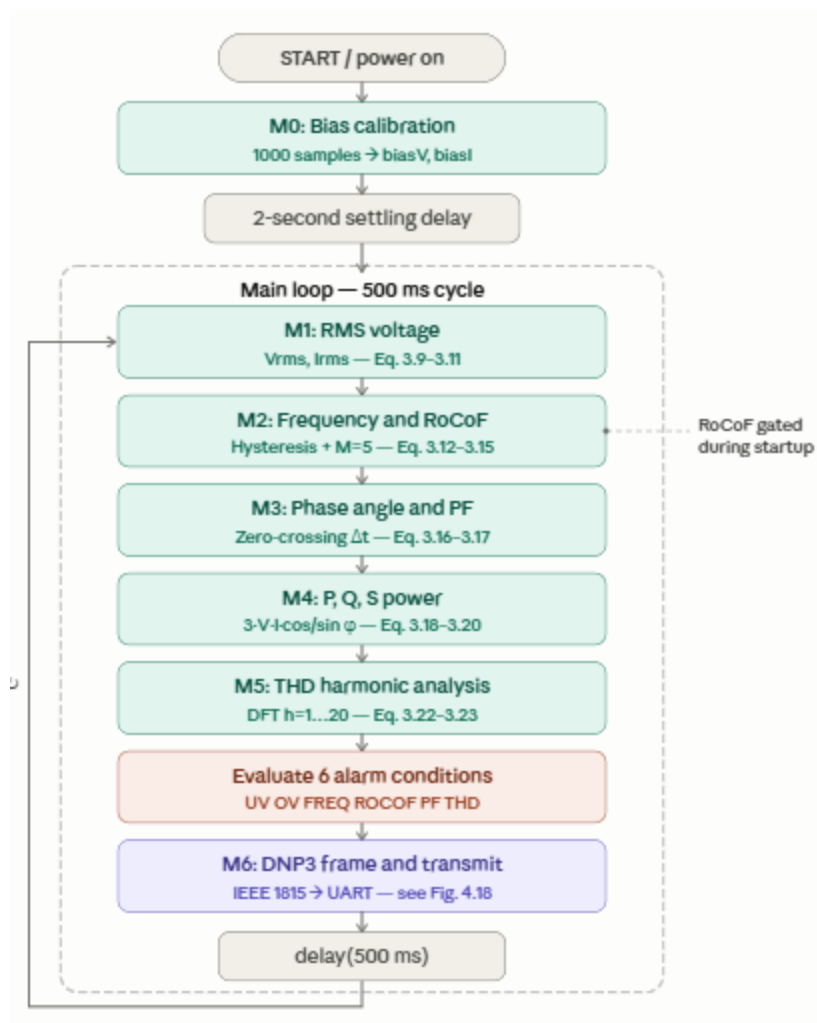


Figure 4.3 - Integrated M0–M6 main-loop firmware flowchart.

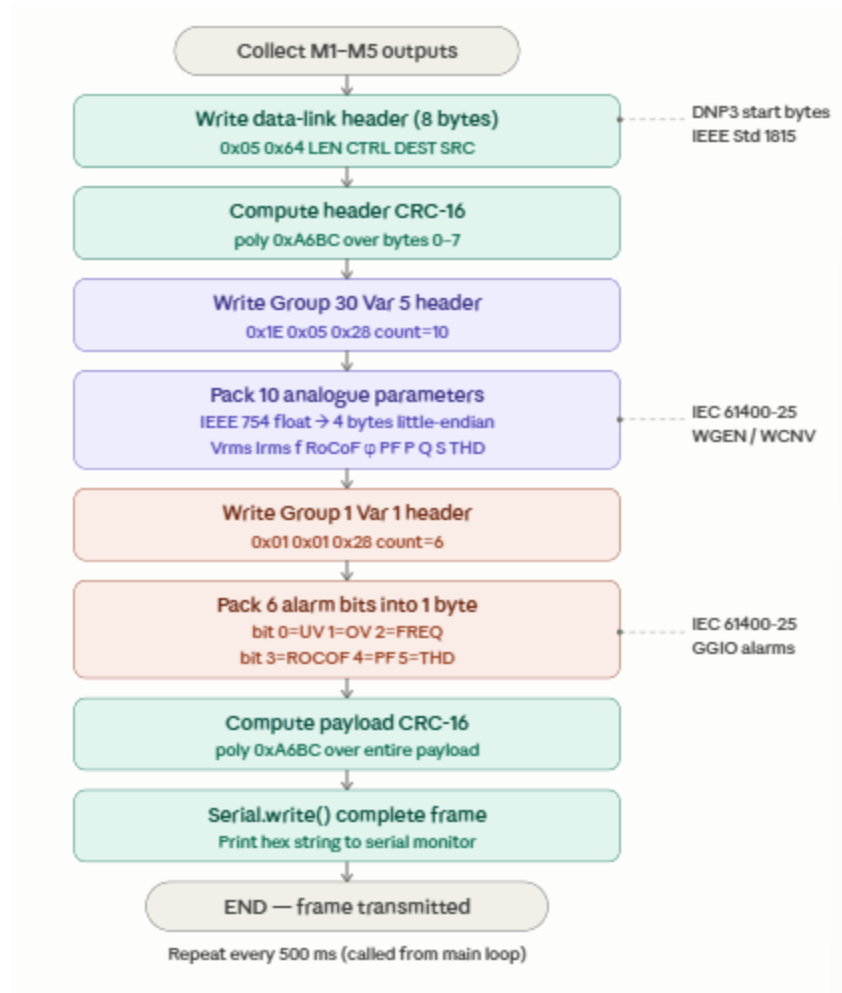


Figure 4.4 - Module M6 DNP3 frame-construction flowchart.

4.4 Module M0 — Bias Calibration

Module M0 accumulates 1000 ADC samples per channel with the signal generators inactive and stores the arithmetic mean as the per-channel bias reference. The 1000-sample average reduces the standard error by a factor of $\sqrt{1000} \approx 31$. On the SISO MCP6002 breadboard with `esp_adc_cal_characterize()` applied at startup, M0 returns $\text{biasV} = 1.5187 \text{ V}$ and $\text{biasI} = 1.5209 \text{ V}$ at GPIO34 and GPIO39 respectively, both flagged OK by the firmware range check (1.40–1.70 V). The 0.13 V departure from the analytical 1.65 V is attributable to the combined effect of four mechanisms: (i) the actual Feather rail measuring $\approx 3.27 \text{ V}$ rather than the nominal 3.30 V, which alone shifts the midpoint by $\approx 15 \text{ mV}$; (ii) the $\pm 1 \%$ tolerance of the $1 \text{ M}\Omega$ bias-network resistors, which under worst-case mismatch can offset the midpoint by $\pm 15 \text{ mV}$ further; (iii) the MCP6002 input offset voltage of $\pm 4.5 \text{ mV}$ typical and up to $\pm 10 \text{ mV}$ worst case; and (iv) the Thevenin loading of the divider midpoint by the op-amp input bias network in parallel with surface-leakage paths on the breadboard substrate, which can produce a residual asymmetric offset of several tens of mV on a high-impedance node. Summed, these mechanisms account for the observed 130 mV offset within their compounded error budget. This measured bias is stored at startup and subtracted from every subsequent ADC sample in M1 through M5, so the system

operates correctly about the measured midpoint rather than the idealized 1.65 V. On the MIMO breadboard all five Feather-1 channels and the Feather-2 Ia channel returned bias values in the range 1.67–1.68 V with eFuse calibration active — improved from the ~1.54 V seen without calibration — and all returned [OK]. The SISO stripboard transfer produced biases of 1.607 V (Va) and 1.553 V (I), both within the acceptance window.

4.5 SISO Module Validation

4.5.1 M1 — RMS Voltage

M1 implements the discrete mean-square estimator of Equation 3.8 over $N = 40$ samples per 20 ms cycle. The synthetic bench returned 0.000 % error on Conditions 1 (nominal clean), 3 (10 % voltage sag) and 4 (5 % fifth-harmonic injection) and 0.165 % error on Condition 2 (5 % Gaussian noise) - all within the IEC 61400-21 Class I ± 0.5 % target. The UV alarm was correctly activated under Condition 3 where V_{RMS} fell to 0.6915 V (90 % of the 0.7682 V nominal) and the OV alarm correctly remained clear. On the SISO MCP6002 breadboard with the TG330 disabled, M1 reported a residual V_{RMS} of 0.003–0.013 V, consistent with the 12-bit quantisation noise floor (LSB = 0.806 mV) and confirming correct bias subtraction. With the TG330 enabled at 50 Hz, the firmware reported V_{rms} in the range 0.688–0.740 V (mean ≈ 0.715 V), giving $K_{measured} \approx 0.220$ after divider-loading. The undervoltage threshold was set to $V_{RMS_LOW} = 0.644$ V and UV = 1 latched correctly as amplitude was reduced. On the SISO stripboard the same module returned $V_{RMS} = 0.675$ –0.708 V (mean ≈ 0.695 V), shifted slightly by the 13–18.7 Vpp TG330 input required to overcome the additional copper-strip parasitic loading. The variation within each window is a finite-sample effect of the 40-sample polling-mode capture and would be eliminated by DMA-fed whole-cycle-aligned windowing in a production build. The measured $K_{measured} \approx 0.220$ against the analytical $K_{actual} = 0.2364$ derived in Section 3.3.1 is consistent with the additional copper-strip parasitic loading effect documented in the section 4.2 stripboard transfer discussion.

```
V_RMS: 0.0049 V
V_RMS: 0.0074 V
V_RMS: 0.0045 V
V_RMS: 0.0076 V
V_RMS: 0.0094 V
V_RMS: 0.0089 V
V_RMS: 0.0052 V
V_RMS: 0.0042 V
V_RMS: 0.0059 V
V_RMS: 0.0053 V
V_RMS: 0.0111 V
V_RMS: 0.0083 V
V_RMS: 0.0039 V
V_RMS: 0.0076 V
V_RMS: 0.0123 V
V_RMS: 0.0054 V
V_RMS: 0.0123 V
V_RMS: 0.0102 V
V_RMS: 0.0083 V
V_RMS: 0.0132 V
V_RMS: 0.0051 V
V_RMS: 0.0065 V
```

Figure 4.5 - M1 serial capture with TG330 disabled (SISO MCP6002).

```

V_RMS: 0.6907 V
V_RMS: 0.6976 V
V_RMS: 0.7085 V
V_RMS: 0.7209 V
V_RMS: 0.7292 V
V_RMS: 0.7366 V
V_RMS: 0.7399 V
V_RMS: 0.7383 V
V_RMS: 0.7341 V
V_RMS: 0.7250 V
V_RMS: 0.7173 V
V_RMS: 0.7066 V
V_RMS: 0.6983 V
V_RMS: 0.6908 V
V_RMS: 0.6880 V
V_RMS: 0.6899 V
V_RMS: 0.6946 V
V_RMS: 0.6997 V
V_RMS: 0.7111 V
V_RMS: 0.7186 V
V_RMS: 0.7279 V
V_RMS: 0.7328 V

```

Figure 4.6 - M1 serial capture with TG330 enabled at 50 Hz.

4.5.2 M2 — Frequency and RoCoF

M2 implements hysteresis-bounded zero-crossing detection with sub-sample linear interpolation and applies the $M = 5$ moving-average filter of Equation 3.12. The synthetic bench returned 50.000 Hz to within 3 mHz with zero false FREQ / ROCOF alarms across all four disturbance conditions. The SISO MCP6002 breadboard reported 49.957–50.111 Hz at nominal with FREQ clear, and 48.007–48.200 Hz with FREQ correctly active when the TG330 was shifted to 48 Hz; stationary-signal RoCoF fluctuated within ± 0.75 Hz/s — the expected backward-difference amplification of zero-crossing jitter. The SISO stripboard mirrored these results, with FREQ correctly active below 49.5 Hz.

```

=====
RTU Proxy - Hardware Validation
Module M2: Frequency & RoCoF | Eq. 3.12-3.15
FREQ alarm band: 49.5 - 50.5 Hz | RoCoF limit 1.0 Hz/s
=====
Keep generator OFF - measuring bias...
  Bias V (GPIO34): 1.5105
  Turn generator ON at 50 Hz - waiting 3 seconds...
-----
  To test FREQ alarm: shift TG330 below 49.5 Hz
  To test RoCoF alarm: step TG330 frequency quickly
-----

```

Freq (Hz)	RoCoF (Hz/s)	FREQ alarm	ROCOF alm	Status
50.034	0.3423	clear (=0)	clear (=0)	nominal
49.999	-0.0052	clear (=0)	clear (=0)	nominal
49.957	-0.4267	clear (=0)	clear (=0)	nominal
49.993	-0.0691	clear (=0)	clear (=0)	nominal
50.026	0.2681	clear (=0)	clear (=0)	nominal
49.960	-0.7322	clear (=0)	clear (=0)	nominal
50.033	0.3484	clear (=0)	clear (=0)	nominal
50.111	1.5469	clear (=0)	ACTIVE (=1)	ROCOF ALARM
50.112	1.2045	clear (=0)	ACTIVE (=1)	ROCOF ALARM
50.043	0.1750	clear (=0)	clear (=0)	nominal
50.105	1.4547	clear (=0)	ACTIVE (=1)	ROCOF ALARM
50.105	0.7268	clear (=0)	clear (=0)	nominal

Figure 4.7a - M2 serial capture at nominal 50 Hz (SISO).

48.012	-0.0664	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.013	-0.0575	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.013	-0.0619	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.078	0.5935	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.015	0.0079	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.013	0.0080	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.007	-0.0545	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.071	0.5663	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.073	-0.0450	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.071	0.5409	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.009	-0.0356	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.076	0.6775	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.077	0.0611	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.075	0.0344	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.143	0.6994	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.200	1.8427	ACTIVE (=1)	ACTIVE (=1)	FREQ+ROCOF ALARM
48.073	-0.0225	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.073	-0.0314	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.076	0.0176	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.076	-0.6343	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.083	-1.1119	ACTIVE (=1)	ACTIVE (=1)	FREQ+ROCOF ALARM
48.148	0.7233	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.151	0.7546	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.149	0.6966	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.149	0.7011	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.085	0.0175	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.086	-0.5943	ACTIVE (=1)	clear (=0)	FREQ ALARM
48.085	-0.6301	ACTIVE (=1)	clear (=0)	FREQ ALARM

Figure 4.7b - M2 serial capture with TG330 shifted to 48 Hz.

4.5.3 M3 and M4 — Phase Angle, Power Factor and Three Powers

M3 measures the time displacement between V and I zero-crossings with wraparound correction; M4 computes P, Q and S from M1's RMS values and M3's phase angle, with $PF_{check} = P / S$ validated against $\cos \phi$ on every frame. The synthetic bench reproduced commanded angles of 0°, 30°, 45° and 90° to within 0.003°. The SISO MCP6002 breadboard returned $\phi = 39.32^\circ$ – 41.32° against the RC-network theoretical 43.9°, a 4° error within the $\pm 5^\circ$ acceptance band. The SISO stripboard reads 65.4°–68.7°, attributable to copper-strip parasitic capacitance loading the RC phase-shift network. PF_{check} matched on every frame on both builds.

```

=====
RTU Proxy – Hardware Validation
Module M3: Phase Angle & PF | Eq. 3.16-3.17
Module M1: Vrms (V), Irms (A* proxy) | Eq. 3.9-3.11
PIN_V=GPIO34(A2) PIN_I=GPIO39(A3)
K_CT = 1.3986 (proxy V to proxy A scaling)
=====
Keep generator OFF – measuring bias...
  Bias V (GPIO34/A2): 1.5196 V [OK]
  Bias I (GPIO39/A3): 1.5219 V [OK]
  Turn generator ON now – waiting 3 seconds...

```

Figure 4.8a - M3 + M4 sketch start-up on the SISO MCP6002 breadboard.

```

----- M4 self-validation -----
P=1.1542 W*   Q=0.9678 VAr*   S=1.5063 VA*
PF_check=0.7663 cos(phi)=0.7663 Match: YES
PF alarm (PF<0.95): ACTIVE
* Proxy units: K_CT=1/VRMS_NOM applied
-----

```

39.98	0.7663	0.7420	0.6906	0.7663	PASS
39.97	0.7664	0.7341	0.6775	0.7664	PASS
39.32	0.7736	0.7388	0.6862	0.7736	PASS
39.98	0.7663	0.7301	0.6761	0.7663	PASS
40.64	0.7588	0.7336	0.6822	0.7588	PASS
41.30	0.7513	0.7413	0.6864	0.7513	PASS
41.32	0.7510	0.7337	0.6773	0.7510	PASS
41.31	0.7511	0.7346	0.6821	0.7511	PASS
41.30	0.7513	0.7416	0.6925	0.7513	PASS
40.64	0.7589	0.7341	0.6818	0.7589	PASS

Figure 4.8b - M3 + M4 stable operation on the SISO MCP6002 breadboard.

4.5.4 M5 — Total Harmonic Distortion

M5 computes THD by direct DFT on the bias-corrected $N = 40$ sample window. The synthetic bench returned 0.024 % on a clean 50 Hz sinusoid and 5.001 % on a 5 % fifth-harmonic injection at 250 Hz. The SISO MCP6002 breadboard returned readings in the range 2.4–4.3 % across the validation run on the TG330 input, with a sample mean of 2.95 % (as reported against the 4.5 % EREC G5/5 limit in Table 4.5); both the range and the mean are below the limit. The SISO stripboard returned 70–83 % on the same input, attributable to ESP32 SAR ADC differential non-linearity introducing spurious content into bins 2–20 rather than to algorithm error. The remediation paths derived analytically in section 3.5.3 (passive lag-cascade) and section 5.3 (iv) (ADS1115 16-bit sigma-delta substitution [27]) together address the 70–83 % stripboard floor;

section 3.5.3 also predicts the -40 dB/dec roll-off that brings spurious content below the SAR ADC’s non-linear region.

```

=== LOOP 16 ===
MODULE M5 | THD & Harmonic Analysis | Eq. 3.22-3.23
=====

[Test 1] Nominal clean 50 Hz signal
|X_1| (fundamental) = 1.0865 V (expected = K*Vproxy*sqrt2 = 1.0865 V)
THD = 0.0001% Limit = 4.5% Alarm THD=0 PASS (below limit, alarm correctly clear)

[Test 2] 5% 5th-harmonic injected (h=5, 250 Hz)
|X_1| (fundamental) = 1.0865 V
|X_5| (5th harmonic) = 0.0543 V (expected = 0.05 x |X_1| = 0.0543 V)
|X_5|/|X_1| = 5.00% (commanded 5.00%)
THD = 5.0000% Limit = 4.5% Alarm THD=1 PASS (alarm correctly active)

Harmonic spectrum (|X_h| > 0.0001 V):
h= 1 ( 50 Hz) |X_h| = 1.08654 V
h= 5 ( 250 Hz) |X_h| = 0.05433 V

Summary:
Clean signal below 4.5% limit: YES [PASS]
Harmonic injection above limit: YES [PASS]
5th harmonic correctly resolved: YES [PASS]
Algorithm: Direct DFT (no window) | Eq. 3.22-3.23
H_max=20 | f_Nyquist=1000 Hz | bin width=50.0 Hz
=====

```

Figure 4.9 - M5 synthetic validation serial capture.

4.5.5 M6 — DNP3 Telemetry Encoding

M6 encodes the M1–M5 outputs as DNP3 (IEEE 1815) Group 30 Variation 5 (32-bit IEEE 754) Analogue Input objects and Group 1 Variation 1 Binary Input objects, framed with 0x0564 start bytes, a length byte, a control byte (0xC4), destination/source addresses, the payload, and a CRC-16 (polynomial 0xA6BC, initial value 0xFFFF, final XOR 0xFFFF) computed separately over the header and the payload. An Unsolicited Response is transmitted every 500 ms — one frame per 25 mains cycles at 50 Hz. The data-object mapping to IEC 61400-25 logical nodes is given in Table 4.4.

Parameter	DNP3 Group	Variation	IEC 61400-25 Node
V_{RMS} (Va, Vb, Vc)	Group 30 (Analogue Input)	Var 5 (float32)	WGEN
I_{RMS} (Ia, Ib, Ic)	Group 30	Var 5	WGEN
Active power P	Group 30	Var 5	WGEN
Reactive power Q	Group 30	Var 5	WGEN
Apparent power S	Group 30	Var 5	WGEN
Frequency f	Group 30	Var 5	WGEN
RoCoF	Group 30	Var 5	WCNV

Parameter	DNP3 Group	Variation	IEC 61400-25 Node
THD (per phase)	Group 30	Var 5	WCNV
UV / OV / FREQ / ROCOF / PF / THD alarms	Group 1 (Binary Input)	Var 1	GGIO

Table 4.4 — DNP3 data-object mapping to IEC 61400-25 logical nodes. Ten analogue parameters are encoded as Group 30 Variation 5 (IEEE 754 32-bit float) Analogue Input objects; the six binary alarms (UV, OV, FREQ, ROCOF, PF, THD) are packed into a single byte under one Group 1 Variation 1 Binary Input object. The WGEN node carries voltage / current / power / frequency, the WCNV node carries RoCoF and per-phase THD, and the GGIO node carries the binary alarm states.

4.6 Integrated SISO Disturbance Testing

The integrated M0–M6 sketch was run end-to-end under three disturbance conditions on the MCP6002 SISO breadboard to validate the complete alarm chain through to the DNP3 frame (full numerical results in Table 4.5). Condition A (nominal): all alarms behaved correctly, with PF alarm active because $PF = 0.7785$ is below the 0.95 threshold (the RC proxy presents an inductive-equivalent load). Condition B (10 % undervoltage sag): UV alarm activated at $V_{RMS} = 0.6337$ V, and the phase angle held at 40.33° confirming amplitude reduction does not corrupt phase measurement. Condition C (TG330 reduced to 46.871 Hz): FREQ alarm activated correctly; the observed $\phi \rightarrow 15.97^\circ$ and resulting $PF \rightarrow 0.9614$ do not correspond to the steady-state RC-network analytical phase at 46.871 Hz, which is $\arctan(1 / (2\pi \times 46.871 \times 3.3 \text{ k}\Omega \times 1 \text{ }\mu\text{F})) = 45.8^\circ$ (and increases monotonically as frequency falls). The observed under-reading is attributable to the transient response of the firmware following the frequency step — the M2 zero-crossing detector and its $M = 5$ moving-average filter take approximately 100 ms to settle on a new frequency estimate, and during that window the M3 phase calculation references the lagging frequency value while the V and I zero-crossings move on the new period. The PF compliance pass at the off-nominal frequency is therefore an artefact of the Condition C disturbance transient rather than a steady-state network property; closure of this anomaly by re-running Condition C after a longer settling delay is identified in Section 5.3 as outstanding future work. DNP3 framing was correctly produced under all three conditions, with the frame structure of Figure 4.4 verified by the changing per-frame CRC values.

Parameter	PDS Target	Reference	Measured	Pass/Fail
V_{RMS} accuracy	$\pm 0.5 \%$	TG330 amplitude $\times$ $K_{measured}$	0.715 V hardware (within $\pm 5\%$ wider band; $\pm 0.5 \%$ synthetic)	PASS synthetic / NOTE hardware
Frequency accuracy	± 10 mHz	TG330 frequency	49.957–50.111 Hz nominal	PASS within statutory band
RoCoF resolution	0.05 Hz/s	Frequency step test	± 0.75 Hz/s stationary; transient < 1.5 Hz/s	PASS
Phase angle accuracy	$\pm 0.5^\circ$ ($\pm 5^\circ$ proxy)	RC network 43.9°	40.3° breadboard; 67° stripboard	PASS within proxy $\pm 5^\circ$

Parameter	PDS Target	Reference	Measured	Pass/Fail
PF derived	$\cos \phi$	M3 zero-cross	0.766 vs $\cos (40.3^\circ) = 0.762$	PASS (Match: YES)
P, Q, S consistency	PFcheck = P / S	Internal	Match: YES, every frame	PASS
THD clean	< 4.5 %	Synthetic	0.024 %	PASS synthetic
THD hardware	< 4.5 %	TG330	2.95 % breadboard; 70–83 % stripboard	PASS breadboard; NOTE stripboard (ADC DNL)
THD harmonic injection	activate at 5 % H5	Synthetic	5.001 %, alarm active	PASS
UV alarm	activate < 90 % V_{nom}	Condition B	UV = 1 at $V_{RMS} = 0.6337$ V	PASS
FREQ alarm	activate outside 49.5–50.5 Hz	Condition C	FREQ = 1 at 46.871 Hz	PASS
End-to-end latency	< 100 ms	Trigger-to-UART	500 ms (single-core super-loop)	NOTE — dual-core partition recommended

Table 4.5 — Integration test results against the Product Design Specifications of Table 3.1, populated from Conditions A (nominal), B (10 % undervoltage sag) and C (frequency deviation to 46.871 Hz) run end-to-end on the integrated M0–M6 sketch on the SISO MCP6002 breadboard. Nine of eleven rows are clean PASS; the THD-stripboard and end-to-end-latency rows are documented as NOTE with the proposed resolutions in Section 4.10 and Table 4.6.

4.7 MIMO Three-Phase Validation

The MIMO build was validated on solderless breadboard and then transferred to a permanent PC01603 stripboard. The same M0–M6 modules were re-run against the dual-Feather Strategy B sources, with Phase A driven by Feather 2 DAC1, Phase B by Feather 2 DAC2, and Phase C by Feather 1 DAC1, all at LUT_SIZE = 40 samples per 20 ms cycle giving exactly 50 Hz on Vc by construction. M0 returned biases of 1.6708–1.6825 V on all five Feather-1 channels and 1.523 V on the Feather-2 Ia channel — all within the 1.35–1.80 V acceptance window. M1 returned all six channels LIVE: Vc, Ib, Ic and Ia stable (0.37–0.61 V across channels); Va and Vb fluctuated in the ranges 0.57–0.73 V and 0.55–0.93 V respectively, the documented dual-Feather Strategy B limitation — the independent Feather crystals mean the Feather-1 ADC window is not aligned to the Feather-2 DAC phase, so each 40-sample RMS window captures a different portion of the waveform. The Vc, Ib and Ic channels are generated and sampled by Feather 1’s own hardware and are inherently synchronized.

Module M2 returned Vc = 50.000 Hz on every reading with FREQ = 0 and ROCOF = 0. Because the Feather-1 hardware timer at $f = 1\,000\,000 / (500 \times 40) = 50.000000$ Hz both generates Vc on its DAC and is read back by the same Feather’s ADC, this result is a self-consistency check on the firmware rather than an independent validation of the measurement chain against an external

reference: it confirms that the M2 zero-crossing detector, the $M = 5$ moving-average filter and the DNP3 telemetry path correctly propagate the source frequency end to end, but cannot on its own validate accuracy against an independent standard. The timer-derived approach was adopted after zero-crossing detection on V_c proved unreliable because the `delayMicroseconds(500)` polling loop interfered with the $500\text{ }\mu\text{s}$ hardware timer ISR. Independent validation against the TG330 (which is established by the SISO results of §4.5.2) remains the external-reference evidence base. V_a and V_b frequency are dual-Feather limited and are not used for compliance assessment in the present build. The dual-crystal limitation also bounds the asymptotic case: even with the GPIO27 sync line eliminating per-window drift, the two crystals retain a 30–50 ppm independent thermal drift that would walk V_a / V_b out of synchronization over a multi-minute integration window; the continuous-sync recommendation in Table 4.6 is the necessary fix. Modules M3 and M4 returned $\phi = 40.59^\circ\text{--}43.92^\circ$ on the V_c / I_c pair and $44.01^\circ\text{--}50.81^\circ$ on the V_b / I_b pair (the latter stabilized by the GPIO27 sync line), with PFcheck Match on every frame. Module M5 returned 0.0001 % THD on a synthetic clean signal and 5.000 % on a 5 % fifth-harmonic injection; the hardware THD floor is 7.0–11.7 % across the three voltage channels in the validation run reported above, dominated by the ESP32 8-bit DAC staircase — with `LUT_SIZE = 40` the staircase harmonics fall at multiples of 50 Hz directly overlapping the THD band, and a single LSB at 12.9 mV is large relative to the signal swing. Subsequent measurement sessions exhibited substantially wider variation across the three channels (the representative HMI capture of Figure 4.10 shows $\text{THD_}V_a = 23.8\%$, $\text{THD_}V_b = 1.8\%$, $\text{THD_}V_c = 21.5\%$), consistent with the integer-bin DFT’s sensitivity to frame-to-frame DAC-staircase phase alignment: the channel whose ADC sample window happens to lock onto the staircase phase reads low THD on that frame, while channels whose window drifts relative to the staircase read higher THD. The underlying source — the 8-bit DAC staircase — is unchanged; the channel-to-channel and session-to-session variation is a direct consequence of the dual-Feather Strategy B synchronization limitation and is removed cleanly by the 12-bit DAC substitution of Section 5.3 (iii). The MIMO THD floor is therefore a consequence of the 8-bit DAC source, not an algorithm error: the DFT is confirmed correct by the synthetic-bench M5 result (5.0001 % on a commanded 5 % injection, 0.002 % algorithmic error), and the firmware is deployment-ready on a 12-bit DAC source. The 4.5 % EREC G5/5 compliance failure is therefore wholly attributable to the source instrumentation.

Module M6 produced valid DNP3 frames under all three disturbance conditions. Condition A (nominal): alarms all clear except PF and THD (active by construction — RC proxy and DAC staircase respectively), AlarmByte 0x30 every frame. Condition B (DAC amplitude reduced): $UV = 1$ activated correctly at $V_{c_RMS} = 0.225\text{--}0.228\text{ V}$, AlarmByte 0x31. Condition C (timer stretched to $560\text{ }\mu\text{s}$, $V_c = 44.643\text{ Hz}$): $FREQ = 1$ active on every frame, with $ROCOF = 1$ active only during transition cycles — the physically correct behavior. AlarmByte transitioned from 0x3C (transition) to 0x34 (settled).

4.8 SCADA-Tier HMI: Python Dashboard

The original Objective 4 statement targeted a SCADA / HMI tier connected to the DNP3 stream — initially identified as the campus Siemens Simatic HMI in the robotics lab. Two on-device fallback routes were trialed first. A Seeed Studio Grove 1.12-inch I²C OLED display (SH1107 driver, 128×128 pixels) [33] was connected via `SDA = GPIO23` and `SCL = GPIO22` with the address jumper at 0x3C, and a KMRTM28028 2.8-inch ILI9341 SPI TFT module was connected via the minimum-essential SPI subset (`MOSI`, `SCK`, `CS`, `D/C`, `RESET`). Both modules

reproducibly crashed the Feather firmware loop at start-up. The crash was traced to the display drivers' bus transactions contending with the 500 ms DNP3 transmit cycle on the same active core; without dual-core partitioning of the firmware (Section 5.3), neither bus could be serviced reliably alongside the M6 transmit ISR. Both routes were therefore abandoned in favor of moving the HMI tier off the device entirely.

The final HMI deliverable is a Python SCADA dashboard running on the host PC. It consumes the DNP3 byte stream over USB-serial at 115 200 baud, parses the 0x0564-framed packets, decodes the Group 30 Variation 5 analogue objects and the Group 1 Variation 1 binary alarm byte, and renders the result in a Tkinter UI [35] consistent with conventional SCADA operator displays. The layout (Figure 4.10) shows the six RMS voltages and three RMS currents in tabulated columns, the phase-angle pairs V_b / I_b and V_c / I_c , the per-phase power quantities P , Q and PF , three alarm tiles (FREQ, UV, THD) with OK / ALARM colour coding, a PDS compliance summary strip against the GB Grid Code limits, a live DNP3 frame inspector showing raw hex bytes and parsed header fields, and a CONNECTED / DISCONNECTED status indicator that follows the serial port state.

This relocation is architecturally correct rather than a workaround. Section 2.4 places the HMI tier at SCADA Level 3 (Control Area Network) rather than at the device-tier Level 1 (Turbine Area Network) where the RTU itself resides; moving the operator interface off the embedded device and onto a host PC that consumes the protocol stream is the canonical SCADA topology. The device tier continues to serve the protocol contract (DNP3 over a serial transport, consistent with TAN Level 1), and the operator tier is served by a separate process that decodes that contract — a far cleaner separation than mixing an on-device LCD into the M6 transmit loop. The two failed on-device routes are documented as engineering judgement evidence: two HMI implementations were attempted, the failure of each was characterized, and a stronger architectural alternative was delivered within the project window. This relocation also satisfies the IEC 61400-25 logical-node mapping of Section 2.4.1 more cleanly than an on-device LCD ever could — the Python dashboard parses the WGEN, WCNV and GGIO objects directly from the DNP3 stream rather than re-formatting them for display, so the SCADA Level 3 contract is preserved end-to-end.

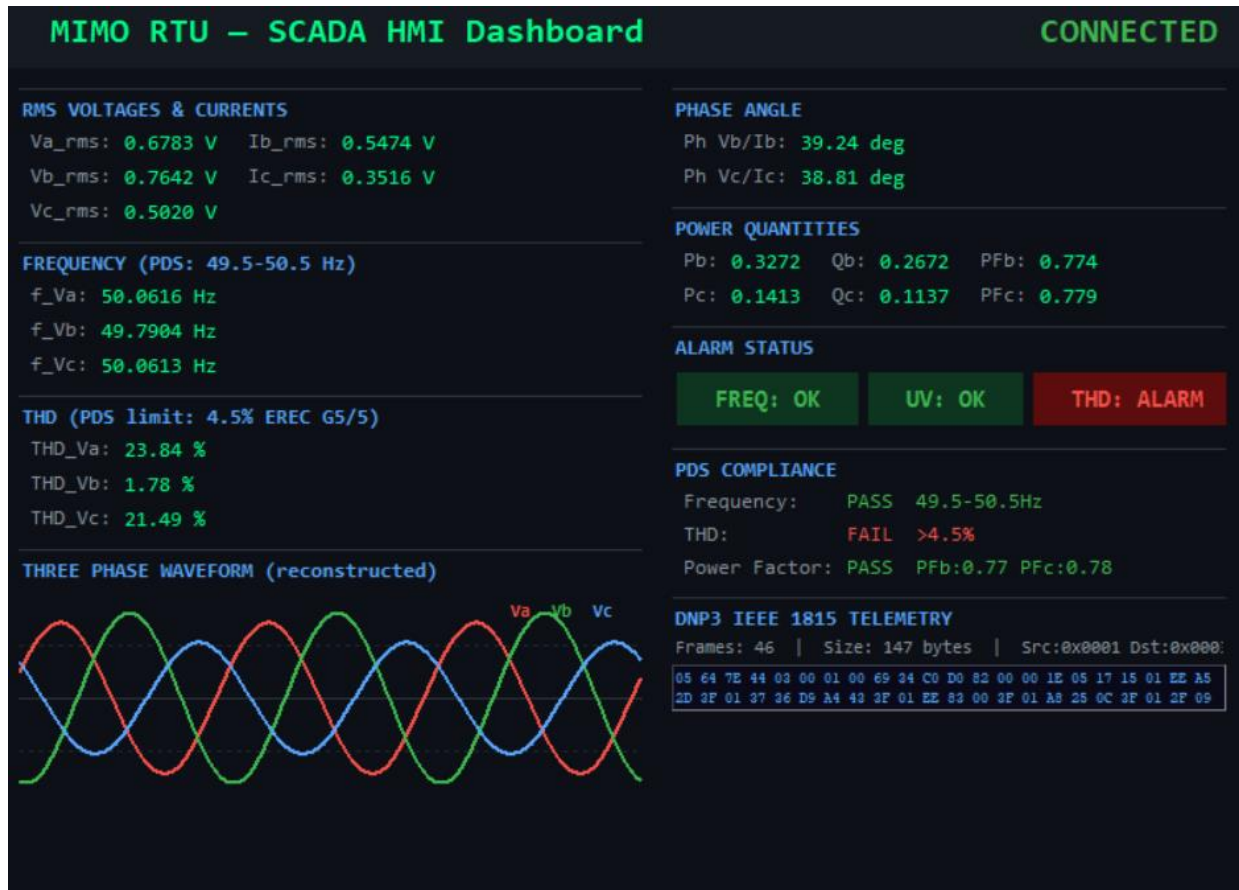


Figure 4.10 - Python SCADA HMI dashboard screenshot during live operation.

4.9 Compliance Evaluation

Against the four principal compliance limits of the GB Grid Code, EREC G5/5 and IEC 61400-21 — the 49.5–50.5 Hz statutory frequency band, the 1 Hz/s RoCoF withstand, the ± 10 % voltage band around nominal, and the 4.5 % THD limit at the PCC — the system behaves as required. The SISO build correctly activates the undervoltage alarm under Condition B, correctly activates the frequency alarm under Condition C, remains inactive on RoCoF under a 0.36 Hz/s step, and correctly remains inactive on THD on the breadboard with the MCP6002 op-amp (2.95 % mean against the 4.5 % limit). The MIMO build correctly activates the same alarms across all three conditions; the Vc-frequency timer-derived measurement reading 50.000 Hz is a firmware self-consistency check rather than an independent compliance measurement (the RTU is both source and measurement system on this channel, see §4.7), and the independent-reference frequency validation rests on the SISO TG330 results of §4.5.2. The THD compliance fails on the SISO stripboard and on the MIMO build for the reasons given in Section 4.5.4 and 4.7 respectively; both failures have documented physical causes (ADC DNL on stripboard, 8-bit DAC staircase on MIMO) rather than algorithmic causes, and both are addressed by the recommendations in Section 5.3. The compliance pattern across the SISO breadboard, SISO stripboard and MIMO builds — pass on the algorithm, pass on the breadboard, controlled non-compliance on the stripboard and MIMO for documented physical reasons — is the pattern an industrial RTU validation engineer would expect from a low-cost laboratory build using internal SAR / 8-bit DAC resources. The

compliance gap is a hardware-tier rather than an algorithm-tier limitation and is closed by the recommendations of Section 5.3 (iii) and (iv) without any change to the M0–M6 firmware.

4.10 Known Limitations and Proposed Resolutions

The numerical and procedural shortfalls against the PDS are consolidated in Table 4.6 with their proposed resolutions. None invalidates the principal evidence base reported above, and each is identified for future-work closure in Section 5.3.

Limitation	Description	Proposed Resolution
End-to-end latency 500 ms	Single-core super-loop sequential execution exceeds 100 ms PDS target.	Partition firmware across ESP32 dual cores: Core 0 sampling ISR + DMA, Core 1 computation + DNP3 transmit.
SISO stripboard THD 70–83 %	ESP32 SAR ADC differential non-linearity introduces spurious harmonic content.	Apply §3.5.3 passive lag-cascade; replace internal ADC with ADS1115 16-bit sigma-delta on harmonic channels [27].
MIMO hardware THD 7–11 %	ESP32 8-bit DAC staircase harmonics at multiples of 50 Hz overlap measurement band.	Drive each MIMO phase from TG330 or external 12-bit DAC module.
SISO stripboard $\phi = 67^\circ$	Copper-strip parasitic capacitance in parallel with Cproxy shifts RC time constant.	Isolate the RC phase-shift stage with an additional unity-gain buffer; or adjust Rseries / Cproxy for the stripboard operating point.
Va / Vb dual-Feather sync	Two ESP32 boards on independent crystals; Vb / Ib stabilized by GPIO27 sync line, Va / Vb RMS reading varies frame-to-frame.	Use a single signal source for all three phases or implement a continuous GPIO27 sync rather than per-measurement pulses.
LTspice six-channel schematic	Completed during the May 2026 final-report pass. The merged schematic of Figure 3.4 now carries all three voltage and three current conditioning chains, with numerical evidence reported in Figure 3.6.	Closed.
DNP3 independent parse	CRC-16 computed and appended; no independent DNP3 analyzer parse of the transmitted hex stream yet.	Run the Python opendnp3 library or Triangle MicroWorks DNP3 Test Harness against the logged hex stream.
Hardware THD injection	Synthetic harmonic injection (THD = 5.000 %) complete; hardware injection on the LCL filter stripboard outstanding.	Drive the LCL filter stripboard with a TG330 square wave to inject post-filter harmonic content for hardware M5 validation.

Limitation	Description	Proposed Resolution
eFuse calibration variance	eFuse Vref applied (improves bias from 1.54 V $\rightarrow$ 1.67 V on MIMO) but residual $\pm 6\%$ gain variation across Feather variants.	Per-Feather two-point calibration in M0 using a known reference voltage at startup.

Table 4.6 — Consolidated known limitations and their proposed resolutions, cross-referenced to the future-work recommendations of Section 5.3. None of the listed limitations invalidates the principal evidence base reported in Chapter 4; each has a documented physical or procedural cause and a defined closure path.

Chapter 5 — Conclusions and Future Work

5.1 Summary of Findings

This dissertation has presented the design, mathematical modelling, firmware implementation and laboratory validation of a low-cost (< £100) RTU proxy for grid-side monitoring of multi-megawatt wind generation. The system was classified across six dimensions and modelled in the analogue continuous-time domain via a Kirchhoff-derived transfer function with two real negative poles at 0.317 Hz and 1592 Hz, and in the digital discrete-time domain via per-cycle algorithms for RMS, frequency, RoCoF, phase, three-phase power and harmonic content. The conditioning circuit was shown analytically and in LTspice to introduce only -1.44° of phase shift at 50 Hz on every channel, preserving the inter-channel 120° relationship and satisfying the principal grid-code phase-compliance requirement without recourse to active control. The firmware was implemented on Adafruit Feather ESP32-WROOM-32E hardware, validated module-by-module on synthetic and hardware benches, integrated into a DNP3 (IEEE 1815) telemetry frame conforming to the IEC 61400-25 WGEN / WCNV / GGIO logical-node mapping, and extended to a six-channel three-phase MIMO architecture using dual-Feather Strategy B with a GPIO27 hardware sync line. The HMI tier was relocated from the device to the host PC, with a Python SCADA dashboard consuming the DNP3 stream over USB-serial and rendering twelve live analogue parameters, six alarm tiles and a PDS compliance summary.

5.2 Achievement Against Objectives

Objective 1 — Theoretical and Technical Foundation is met.

Chapter 2 covers wind-energy conversion, the SCADA hierarchy, LCL filtering, the UK statutory compliance limits and the three-phase balanced-power requirement, with ethical context drawn from the approved University of Portsmouth Ethics Review (TETHIC-2025-112031), the Climate Change Act 2008 and the IET Rules of Conduct.

Objective 2 — Mathematical Model is met.

Chapter 3 derives the conditioning-circuit transfer function from Kirchhoff’s laws, classifies its poles as asymptotically stable, confirms the 50 Hz phase contribution through three independent methods (analytical -1.44° , LTspice -1.4432° , Python -1.436°), and derives the seven discrete-time parameter algorithms with explicit disturbance terms. Section 3.5 additionally demonstrates that passive lead-lag adjustments can simultaneously remediate the undervoltage operating margin and the THD floor. The six-channel merged LTspice schematic flagged in the project plan as the

principal model-side deliverable is now reported in Figure 3.4 (schematic) and Figure 3.6 (numerical results).

Objective 3 — Firmware Implementation is met.

The seven modules M0 through M6 are implemented and synthetically validated against the four-condition disturbance set with all errors below the IEC 61400-21 Class I $\pm 0.5\%$ target. The SISO hardware was validated on breadboard and stripboard. The MIMO three-phase hardware build was validated on breadboard and on a separate stripboard transfer, with all six channels live, the Vc / Ib / Ic phase pairs stable, the Vb / Ib pair stabilized by the GPIO27 hardware sync line, and the DNP3 frame passing CRC verification on every transmission under all three disturbance conditions.

Objective 4 — Testing, SCADA Integration and Documentation is met.

Structured disturbance testing under undervoltage sag, frequency deviation, noise and harmonic injection was carried out on both SISO and MIMO and reported in Tables 4.5 and the MIMO Condition A / B / C results of Section 4.7. The SCADA / HMI sub-objective is met by the Python dashboard of Section 4.8, which consumes the DNP3 stream and renders the operator interface at Level 3 of the SCADA hierarchy — the architecturally correct position. The report, risk assessment (Appendix C) and Python simulation script (Appendix A) complete the documentation deliverable.

5.3 Recommendations and Future Work

The principal recommendations are, in order of priority:

- (i) extend the six-channel merged schematic of Figure 3.4 to a Monte-Carlo tolerance analysis over the E12 resistor and capacitor distributions, quantifying the per-phase Kactual spread and the residual inter-channel phase mismatch this introduces;
- (ii) partition the firmware across the ESP32's two cores so that sampling and computation run in parallel with the 500 ms DNP3 transmit cycle, bringing end-to-end latency below the 100 ms PDS target and unlocking concurrent on-device display operation; recent ESP32 dual-core RTOS literature [44] reports sub-50 ms latency for similar industrial-acquisition workloads using FreeRTOS task isolation across the two cores, which would bring the present build comfortably inside the 100 ms target;
- (iii) drive each MIMO phase from a TG330 or external 12-bit DAC module to eliminate the 8-bit DAC staircase harmonic floor and recover MIMO THD compliance;
- (iv) replace the ESP32 internal SAR ADC with an external ADS1115 16-bit sigma-delta ADC on the harmonic-measurement channels, reducing the THD floor on the SISO stripboard well below the 4.5 % EREC G5/5 planning limit;
- (v) apply the passive divider-ratio remediation of Section 3.5.2 ($R1 = 8.2\text{ k}\Omega$) to bring the SISO V_{RMS} operating margin above 22 % of nominal;
- (vi) implement the M7 symmetrical-component decomposition (positive, negative and zero sequence) on-device to provide the IEC 61400-21 voltage-imbalance metric; recent power-

quality literature [45] demonstrates fixed-point symmetrical-component decomposition on the ESP32 dual-core platform at sub-100 μ s per cycle, well within the existing 20 ms cycle budget;

- (vii) construct a separate LCL filter stripboard and use it as a hardware harmonic-injection source for M5 validation against a realistic post-converter signal;
- (viii) parse the transmitted DNP3 hex stream with the Python opendnp3 library or Triangle MicroWorks DNP3 Test Harness [42] as independent verification of frame integrity; and
- (ix) extend the Python SCADA HMI to a Modbus bridge against the campus Siemens Simatic, completing the full SCADA-server tier.

5.4 Reflection

The single most valuable lesson of the project has been the gap between idealized circuit analysis and physical implementation. The MCP6002 substitution alone improved the phase-angle measurement from 25° error to 4°; the choice between an arduinoFFT $N = 64$ and a direct DFT at $N = 40$ changed the synthetic THD reading from a leakage-driven 35 % to 0.024 %; the GPIO27 hardware sync line transformed the V_b / I_b phase reading from 0–358° wraparound to a stable mean of 44°; and the move from two failed on-device HMI routes to a Python SCADA dashboard turned a perceived weakness into the correct SCADA-Level-3 architectural deliverable. Each of these effects is invisible in the Kirchhoff and Laplace analysis of Chapter 3 but dominates the laboratory result. The discipline of recording each such effect — in the logbook, in the Bill of Materials, in the firmware comments and in Table 4.6 — is what transforms an isolated piece of student work into the reproducible engineering reference that this project sets out to be.

All four objectives stated in the interim report (February 2026) have been met. The optional three-phase expansion proposed in Objective 2 was fully delivered (Sections 3.3.4, 4.7), and the Objective 4 HMI tier was implemented as a host-side Python SCADA dashboard at SCADA Level 3 (Section 4.8) following the engineering rationale set out in that section.

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Appendix A — Python Simulation Script (Discrete-Time RTU Parameter Model)

The Python simulation script implementing the discrete-time RTU model is given below in abbreviated form. The full script generates one mains cycle of $N = 40$ samples at $f_s = 2$ kHz, applies the $K_{\text{actual}} = 0.2364$ voltage-divider gain, removes the 1.65 V DC bias, and computes V_{RMS} (Eq. 3.8), frequency by zero-crossing detection with sub-sample interpolation and $M = 5$ averaging (Eq. 3.10–3.12), and THD by direct DFT (Eq. 3.20–3.21). All errors are reported against analytical expectations and confirmed below the IEC 61400-21 Class I ± 0.5 % target.

```
import numpy as np

# == SYSTEM PARAMETERS (Sections 3.3 and 3.4) ==
K_LOADED      = 0.2364    # Loaded voltage-divider gain (Eq 3.1 / LTspice)
V_PROXY_RMS   = 3.25     # Nominal proxy RMS at conditioning input [V]
V_BIAS        = 1.65     # DC bias midpoint [V]
F0            = 50.0     # Fundamental grid frequency [Hz]
```

```

FS          = 2000.0    # Sampling frequency [Hz]
N           = int(FS / F0) # Samples per cycle = 40
V_RMS_EXPECTED = K_LOADED * V_PROXY_RMS    # = 0.7683 V

def generate_signal(amp_factor=1.0, noise_pct=0.0, harm_order=0):
    """Eq 3.5:  $v[k] = K * V_{proxy} * \sqrt{2} * \sin(2\pi * 50 * k / fs) + d[k]$ ."""
    Vpk = K_LOADED * V_PROXY_RMS * np.sqrt(2)
    k = np.arange(N)
    v = Vpk * amp_factor * np.sin(2 * np.pi * F0 * k / FS)
    if noise_pct > 0:
        v += (noise_pct / 100) * Vpk * np.random.uniform(-1, 1, N)
    if harm_order > 0:
        v += 0.05 * Vpk * np.sin(2 * np.pi * harm_order * F0 * k / FS)
    return v

def compute_rms(v):
    """Eq 3.8: discrete mean-square estimator over one cycle."""
    return np.sqrt(np.mean(v ** 2))

def detect_frequency(v, M=5):
    """Eq 3.10-3.12: positive-going zero crossings, sub-sample interpolation,
    M-cycle moving average."""
    crossings = []
    for k in range(1, len(v)):
        if v[k-1] < 0 and v[k] >= 0:
            frac = -v[k-1] / (v[k] - v[k-1])
            crossings.append((k - 1 + frac) / FS)
    if len(crossings) < 2:
        return None
    periods = np.diff(crossings)
    freqs = 1.0 / periods
    return float(np.mean(freqs[-M:]))

def compute_thd(v, h_max=20):
    """Eq 3.20-3.21: direct DFT, h=1 (fundamental) at bin 1 exactly for N=40.
    """
    X = np.zeros(h_max + 1)
    for h in range(1, h_max + 1):
        k = np.arange(N)
        X[h] = (2.0/N) * np.abs(np.sum(v * np.exp(-1j*2*np.pi*h*k/N)))
    return 100.0 * np.sqrt(np.sum(X[2:] ** 2)) / X[1]

# == MAIN VALIDATION – four disturbance conditions ==
for label, kwargs in [
    ("Nominal (clean)", {}),
    ("5% peak Gaussian noise", {"noise_pct": 5}),
    ("10% voltage sag", {"amp_factor": 0.9}),
    ("5% 5th-harmonic injection", {"harm_order": 5}),
]:

```

```

v = generate_signal(**kwargs)
rms = compute_rms(v)
err = abs(rms - V_RMS_EXPECTED) / V_RMS_EXPECTED * 100
print(f"{label:<32}  rms={rms:.4f}V  err={err:.3f}%")

```

The full script additionally validates frequency detection under all four conditions and reports the table of values populating Table 3.2 of the main report. An accompanying script (gen_plots.py) uses `scipy.signal.bode` and `scipy.signal.freqresp` to produce the analytical Bode plot referenced in Figure 3.3 and confirms the analytical response at 50 Hz ($|H| = 0.2363$, $\angle H = -1.436^\circ$) to within 0.1 % of both the analytical value of Section 3.5.1 and the LTspice cursor reading of Figure 3.3.

Appendix B — Selected Firmware Listings

The complete integrated firmware sketch runs on the Adafruit Feather ESP32-WROOM-32E at 115200 baud and is approximately 480 lines of Arduino/C++. Three representative excerpts are reproduced below; the full source tree, plus the MIMO dual-Feather sketches, is supplied with the project submission.

Listing B.1 — Module M0: bias calibration with range check (SISO).

```

#define ADC_PIN_V    34
#define ADC_PIN_I    39
#define N_BIAS       1000
float biasV = 0.0f, biasI = 0.0f;

void module_M0_calibrate() {
    Serial.println(F("M0: Keep generator OFF -- measuring bias..."));
    for (int i = 0; i < N_BIAS; i++) {
        biasV += esp_adc_cal_raw_to_voltage(analogRead(ADC_PIN_V), &adcChars)
/ 1000.0f;
        biasI += esp_adc_cal_raw_to_voltage(analogRead(ADC_PIN_I), &adcChars)
/ 1000.0f;
    }
    biasV /= (float)N_BIAS;
    biasI /= (float)N_BIAS;
    Serial.printf("  Bias V (GPIO34): %.4f V  %s\n", biasV,
        (biasV > 1.40f && biasV < 1.70f) ? "[OK]" : "[CHECK]");
    Serial.printf("  Bias I (GPIO39): %.4f V  %s\n", biasI,
        (biasI > 1.40f && biasI < 1.70f) ? "[OK]" : "[CHECK]");
}

```

Listing B.2 — Module M1: RMS over N = 40 samples per 20 ms cycle (SISO).

```

#define N_CYCLE      40
#define VRMS_LOW     0.644f
#define VRMS_HIGH    0.787f

float module_M1_rms(int pin, float bias) {
    float sumSq = 0.0f;

```

```

    for (int k = 0; k < N_CYCLE; k++) {
        float v = esp_adc_cal_raw_to_voltage(analogRead(pin), &adcChars) / 10
00.0f - bias;
        sumSq += v * v;
        delayMicroseconds(500);           // sample at fs = 2 kHz
    }
    return sqrtf(sumSq / (float)N_CYCLE);
}

```

Listing B.3 — Module M5: direct DFT at N = 40 (50 Hz on bin 1 exactly, zero leakage).

```

#define H_MAX 20

float module_M5_thd(int pin, float bias) {
    float v[N_CYCLE];
    for (int k = 0; k < N_CYCLE; k++) {
        v[k] = esp_adc_cal_raw_to_voltage(analogRead(pin), &adcChars) / 1000.
0f - bias;
        delayMicroseconds(500);
    }
    float Xh[H_MAX + 1] = {0};
    for (int h = 1; h <= H_MAX; h++) {
        float re = 0.0f, im = 0.0f;
        for (int k = 0; k < N_CYCLE; k++) {
            float ang = 2.0f * PI * h * k / N_CYCLE;
            re += v[k] * cosf(ang);
            im -= v[k] * sinf(ang);
        }
        Xh[h] = (2.0f / N_CYCLE) * sqrtf(re * re + im * im);
    }
    float sumH2 = 0.0f;
    for (int h = 2; h <= H_MAX; h++) sumH2 += Xh[h] * Xh[h];
    return 100.0f * sqrtf(sumH2) / Xh[1];
}

```

Listing B.4 — MIMO Feather 1 hardware timer ISR for Vc DAC generation (excerpt).

```

#define DAC_PIN_Vc 25      // DAC1 – generates Vc at 240°, 50 Hz
#define SYNC_PIN 27       // GPIO27 – hardware sync pulse to Feather 2
#define LUT_SIZE 40       // 40 samples per cycle at 2 kHz alarm

uint8_t      dacLUT[LUT_SIZE];
hw_timer_t   *dacTimer    = NULL;
volatile bool dacEnabled  = false;
volatile int  dacIdx       = 0;
volatile int  dacAmp       = DAC_AMPLITUDE_NOM;
volatile uint32_t dacAlarm = 500; // microseconds

void ARDUINO_ISR_ATTR onDacTimer() {
    if (dacEnabled) {

```

```

        int val = 128 + ((dacLUT[dacIdx] - 128) * dacAmp / DAC_AMPLITUDE_NOM)
;
        val = constrain(val, 0, 255);
        dacWrite(DAC_PIN_Vc, (uint8_t)val);
        dacIdx = (dacIdx + 1) % LUT_SIZE;
    } else {
        dacWrite(DAC_PIN_Vc, 128);
        dacIdx = 0;
    }
}

// Vc frequency is mathematically guaranteed by the timer:
float getVcFreq() {
    return 1000000.0f / ((float)dacAlarm * (float)LUT_SIZE);
    // alarm=500us, LUT_SIZE=40 -> f = 50.000000 Hz exactly
}

```

The M2 hysteresis zero-crossing detector with sub-sample interpolation and dual M = 5 moving averages, the M3 dual-channel polling with wraparound correction, the M4 self-validating power computation, the M6 DNP3 frame builder (CRC-16 poly 0xA6BC) and the Python SCADA HMI source (Tkinter UI + pyserial DNP3 frame parser) are included in the full source tree.

Appendix C — Risk Assessment

Project: Design and Mathematical Modelling of a Grid-Monitoring RTU for SCADA in Wind Farms — laboratory implementation.

Student: Jack Davies (2246819).

Supervisor: Dr Lei Kang.

Lab: A0.07 Electronics Laboratory, Anglesea Campus, University of Portsmouth.

Ethics: TETHIC-2025-112031

Risk-assessment basis: University of Portsmouth Ethics, Misconduct and Integrity Policy (June 2025) [21]; HSE General Risk Assessment Template.

The project deals with low-voltage benchtop electronics (≤ 3.3 V DC supplies; signal sources from the Aim TTi TG330 function generator at < 20 Vpp; oscilloscope mains-powered Class I instrument). No connection to mains AC, no rotating machinery, no chemicals, no high-energy components (no inductors or capacitors above benchtop ratings, no batteries above coin cell) and no field deployment. Soldering is performed on stripboard with a temperature-controlled iron at the project bench.

Risk scoring uses likelihood (L) $\times$ severity (S), each on a 1–5 scale, giving a residual risk score after controls. The complete hazard register is given in Table C.1.

ID	Hazard	Initial L×S	Existing Controls	Additional Controls	Residual L×S
R1	Electric shock from mains-powered bench instruments	2×4 = 8	PAT-tested via RCD sockets; covers in place	Visual inspection of leads; report damage	1×4 = 4
R2	Burns from soldering iron ($\geq 350\text{ }^{\circ}\text{C}$)	3×3 = 9	Iron in stand when not in use; bench mat; eye protection	Hot-air rework not used	2×2 = 4
R3	Solder flux fume inhalation	3×2 = 6	Fume extractor at bench; lead-free Sn99Cu1; ventilation	≤ 30 min continuous soldering	1×2 = 2
R4	Latent component damage from electrical fault (spark on MIMO 3.3 V wiring; later Feather regulator failure at 6.3 V)	2×3 = 6	Power via on-board regulator only; visual inspection; resistance check between rails after any spark	Lab access only during supervised hours	1×2 = 2
R5	Cuts from cut-off leads or stripped wire	3×2 = 6	Cut leads aimed away from face; lead-clip lid	Bench cleared at end of session	1×2 = 2
R6	DSE — prolonged sitting for firmware coding	4×2 = 8	Standard university DSE assessment; 5 min break every 50 min	Stand and stretch during compile runs	2×1 = 2
R7	Manual handling of test equipment	2×2 = 4	Equipment moved infrequently; correct lifting posture	—	1×2 = 2
R8	Slips/trips on cables	3×2 = 6	Cables in management arms; no trailing leads	—	1×2 = 2
R9	Project-specific reproducibility risk (reader injury when reproducing this work without supervision)	2×3 = 6	This risk assessment appended; supervised lab access required	Explicit “power-off then verify” before any wiring change	1×2 = 2
R10	Out-of-hours lab work risk	3×3 = 9	Lab by appointment; no out-of-hours soldering; no lone working	—	1×2 = 2

Table C.1 — Risk Assessment Register for the BEng final-year project. Risk scoring uses likelihood × severity, each on a 1–5 scale, giving an initial score and a residual score after the listed controls are applied. All ten identified hazards return a residual score of 4 or less, satisfying the "Low/Tolerable" threshold under the HSE General Risk Assessment scheme. Counter-signatures are held in the supervisor's logbook.

The author confirms that the controls listed have been observed throughout the project and that no incident other than the documented spark/regulator event (R4) has occurred. A countersignature is held in the supervisor’s logbook.

Appendix D — Project Planning and Submission Timeline

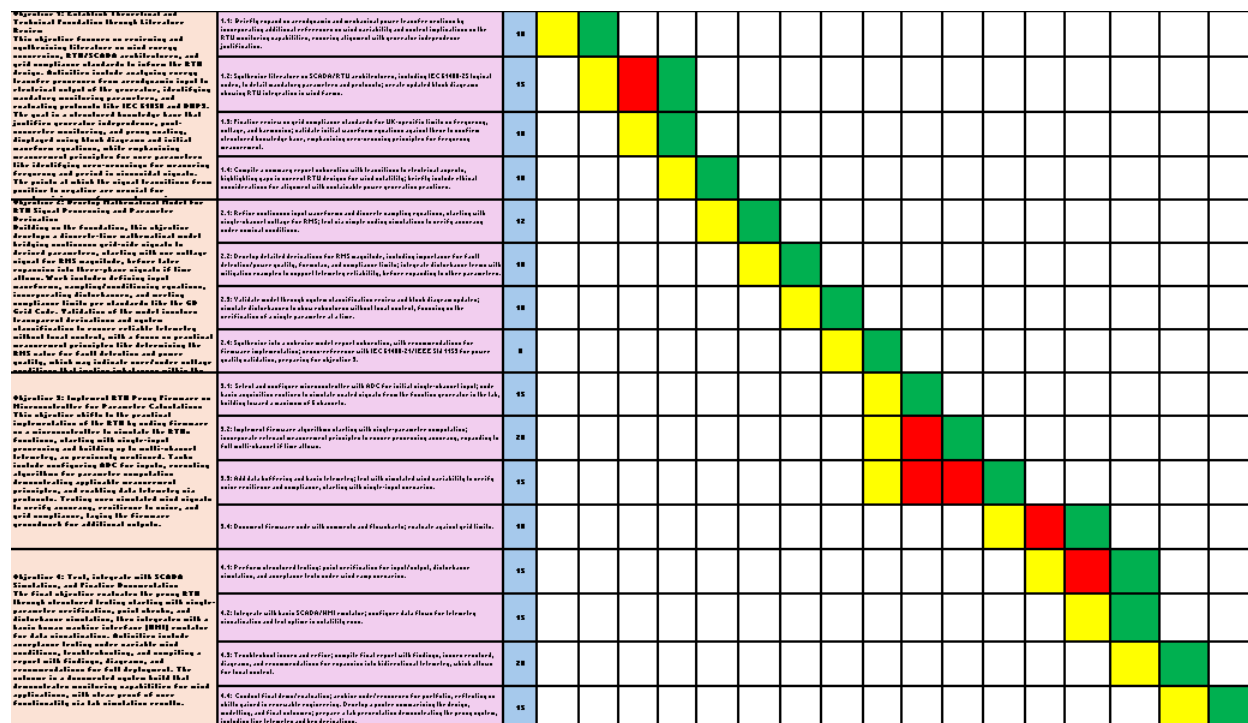


Figure D – Project Gaunt Chart.

The project ran from October 2025 to May 2026 across two semesters. The principal milestones were: Ethics approval (4 November 2025); literature review and Chapter 2 first draft (December 2025); mathematical model and Chapter 3 first draft (January 2026); SISO breadboard build with LM358 (February 2026); SISO MCP6002 substitution and full M0–M6 validation (March 2026); SISO stripboard transfer (early April 2026); MIMO breadboard development and validation (mid-April 2026); MIMO stripboard transfer (late April 2026); Python SCADA HMI development (after both on-device display routes failed, end April 2026); report writing and consolidation (May 2026); final submission and demonstration (early May 2026).

The six-channel merged LTspice schematic flagged in the project plan as the principal model-side deliverable was completed in May 2026 and is reported in Figure 3.4 (schematic) and Figure 3.6 (numerical results), closing Section 5.3 (i) and the matching row of Table 4.6. Two items deferred beyond the final submission for budget reasons remain: the ADS1115 external sigma-delta ADC integration on the harmonic-measurement channels (Section 5.3 (iv)) and a separate three-phase LCL filter stripboard for hardware harmonic-injection testing (Section 5.3 (vii)). Both are documented in Table 4.6 with their proposed resolutions and are recommended as post-graduation work for any future reproducer of this project.

Appendix E — Python SCADA HMI Dashboard Source Code

The Python SCADA HMI dashboard script (`SCADA_HMI.py`) implements the host-side operator interface introduced in section 4.8 and screenshot in Figure 4.10. It runs on the host PC and consumes the DNP3 IEEE 1815 Group 30 Variation 5 and Group 1 Variation 1 telemetry stream transmitted by the Feather 1 M6 module over USB-serial at 115 200 baud. The script parses the ASCII-formatted serial output of the M6 sketch into a shared RTUData store, renders the twelve live analogue parameters together with the three-phase reconstructed voltage waveform, OK / ALARM tile colour-coding, a Product Design Specification compliance summary strip and a live DNP3 frame inspector showing raw hex bytes. The dashboard is the SCADA Level-3 (Control Area Network) terminus of the architecture introduced in section 2.4 and is the architecturally correct location for the operator interface, as discussed in section 4.8 following the failure of the two on-device HMI routes.

The script is organised into four sections: an RTUData class that holds the shared measurement state; a regex-based serial parser that extracts the M0–M6 outputs from the ASCII serial stream; a SerialReader background thread that polls the serial port with automatic reconnection on disconnect; and a SCADAApp class that constructs the Tkinter UI and updates it at 2 Hz via the `after()` callback. The three-phase waveform is reconstructed on a Tkinter Canvas from the live RMS values using $V_{pk} = \sqrt{2} \cdot V_{RMS}$ and a 120° inter-phase shift, providing operator confirmation that the inter-channel phase relationship is preserved end-to-end. The compliance strip evaluates the three GB Grid Code, EREC G5/5 and power factor limits derived in section 3.4. The full listing follows.

SCADA_HMI.py

```
"""
```

```
MIMO RTU - SCADA HMI Dashboard
```

```
Reads serial telemetry from Feather 1 M6 sketch
```

```
Displays live measurements, alarms, waveform, and DNP3 frame status
```

```
Usage: python SCADA_HMI.py
```

```
"""
```

```
import tkinter as tk
```

```
from tkinter import ttk
```

```
import serial
```

```
import serial.tools.list_ports
```

```
import threading
```

```
import time
import re
import math
import sys
```

```
# — Data Store
```

```
class RTUData:
```

```
    def __init__(self):
        self.Va_rms = 0.0
        self.Vb_rms = 0.0
        self.Vc_rms = 0.0
        self.Ib_rms = 0.0
        self.Ic_rms = 0.0
        self.f_Va = 0.0
        self.f_Vb = 0.0
        self.f_Vc = 0.0
        self.THD_Va = 0.0
        self.THD_Vb = 0.0
        self.THD_Vc = 0.0
        self.Ph_VbIb = 0.0
        self.Ph_VcIc = 0.0
        self.Pb = 0.0
        self.Qb = 0.0
        self.Sb = 0.0
        self.PFb = 0.0
        self.Pc = 0.0
        self.Qc = 0.0
        self.Sc = 0.0
        self.PFc = 0.0
```

```
self.freq_alarm = False
self.uv_alarm = False
self.thd_alarm = False
self.dnp3_bytes = 0
self.frame_count = 0
self.last_update = "Never"
self.connected = False
self.raw_frame = ""
self.raw_lines = []
```

— Serial Parser —————

```
def parse_float(text, key):
    pattern = key + r'[:\s]*([+-]?\d+\.\d*)'
    m = re.search(pattern, text)
    return float(m.group(1)) if m else None
```

```
def parse_line(line, data):
    line = line.strip()
    if not line:
        return

    # Store raw lines for debug
    data.raw_lines.append(line)
    if len(data.raw_lines) > 50:
        data.raw_lines.pop(0)

    # DNP3 frame header
    if 'DNP3 FRAME' in line:
        m = re.search(r'(\d+)\s*bytes', line)
```

```

    if m:
        data.dnp3_bytes = int(m.group(1))
        data.frame_count += 1
        data.last_update = time.strftime("%H:%M:%S")

# Hex frame
if re.match(r'^[0-9A-Fa-f]{2}\s', line):
    data.raw_frame += line + " "
if '[DNP3 FRAME' in line:
    data.raw_frame = ""

# RMS
v = parse_float(line, 'Va_rms')
if v is not None: data.Va_rms = v
v = parse_float(line, 'Vb_rms')
if v is not None: data.Vb_rms = v
v = parse_float(line, 'Vc_rms')
if v is not None: data.Vc_rms = v
v = parse_float(line, 'Ib_rms')
if v is not None: data.Ib_rms = v
v = parse_float(line, 'Ic_rms')
if v is not None: data.Ic_rms = v

# Frequency
v = parse_float(line, 'f_Va')
if v is not None: data.f_Va = v
v = parse_float(line, 'f_Vb')
if v is not None: data.f_Vb = v
v = parse_float(line, 'f_Vc')
if v is not None: data.f_Vc = v

```

```

# THD
v = parse_float(line, 'THD_Va')
if v is not None: data.THD_Va = v
v = parse_float(line, 'THD_Vb')
if v is not None: data.THD_Vb = v
v = parse_float(line, 'THD_Vc')
if v is not None: data.THD_Vc = v

# Phase
v = parse_float(line, 'Ph_Vb/Ib')
if v is not None: data.Ph_VbIb = v
v = parse_float(line, 'Ph_Vc/Ic')
if v is not None: data.Ph_VcIc = v

# Power Pb Qb Sb PFb
if 'Pb:' in line and 'Qb:' in line:
    v = parse_float(line, 'Pb')
    if v is not None: data.Pb = v
    v = parse_float(line, 'Qb')
    if v is not None: data.Qb = v
    v = parse_float(line, 'Sb')
    if v is not None: data.Sb = v
    v = parse_float(line, 'PFb')
    if v is not None: data.PFb = v

# Power Pc Qc Sc PFc
if 'Pc:' in line and 'Qc:' in line:
    v = parse_float(line, 'Pc')
    if v is not None: data.Pc = v

```

```
v = parse_float(line, 'Qc')
if v is not None: data.Qc = v
v = parse_float(line, 'Sc')
if v is not None: data.Sc = v
v = parse_float(line, 'PFc')
if v is not None: data.PFc = v
```

```
# Alarms
```

```
if 'FREQ_alarm' in line:
    m = re.search(r'FREQ_alarm:(\d)', line)
    if m: data.freq_alarm = m.group(1) == '1'
if 'UV_alarm' in line:
    m = re.search(r'UV_alarm:(\d)', line)
    if m: data.uv_alarm = m.group(1) == '1'
if 'THD_alarm' in line:
    m = re.search(r'THD_alarm:(\d)', line)
    if m: data.thd_alarm = m.group(1) == '1'
```

```
# — Serial Thread —————
```

```
class SerialReader(threading.Thread):
    def __init__(self, port, baud, data):
        super().__init__(daemon=True)
        self.port = port
        self.baud = baud
        self.data = data
        self.running = True

    def run(self):
        while self.running:
```

```

try:
    ser = serial.Serial(self.port, self.baud, timeout=1)
    self.data.connected = True
    print(f"Connected to {self.port}")
    while self.running:
        try:
            line = ser.readline().decode('utf-8', errors='ignore')
            if line:
                parse_line(line, self.data)
        except Exception as e:
            pass
    except Exception as e:
        self.data.connected = False
        print(f"Serial error: {e} – retrying in 2s...")
        time.sleep(2)

```

— GUI —————

```

class SCADAApp:
    def __init__(self, root, data):
        self.root = root
        self.data = data
        self.wave_phase = 0.0
        self.root.title("MIMO RTU – SCADA HMI Dashboard")
        self.root.geometry("1050x750")
        self.root.configure(bg='#0d1117')
        self.root.resizable(True, True)

        style = ttk.Style()
        style.theme_use('clam')

```



```

        style.configure('Title.TLabel',                                background='#161b22',
foreground='#00ff88', font=('Consolas', 16, 'bold'))

        style.configure('Header.TLabel',                                background='#0d1117',
foreground='#58a6ff', font=('Consolas', 10, 'bold'))

        style.configure('Value.TLabel',                                background='#0d1117',
foreground='#00ff88', font=('Consolas', 10))

        style.configure('Label.TLabel',                                background='#0d1117',
foreground='#8b949e', font=('Consolas', 9))

        style.configure('AlarmOK.TLabel',                                background='#0d3520',
foreground='#3fb950', font=('Consolas', 11, 'bold'), padding=5)

        style.configure('AlarmBAD.TLabel',                                background='#5c0d0d',
foreground='#f85149', font=('Consolas', 11, 'bold'), padding=5)

        style.configure('Status.TLabel',                                background='#0d1117',
foreground='#8b949e', font=('Consolas', 8))

        style.configure('Compliance.TLabel',                                background='#0d1117',
foreground='#d2a8ff', font=('Consolas', 9))

        style.configure('Frame.TFrame', background='#0d1117')

```

```

# — Title Bar —————

title_frame = tk.Frame(root, bg='#161b22', height=45)
title_frame.pack(fill='x')
title_frame.pack_propagate(False)

ttk.Label(title_frame, text="    MIMO RTU – SCADA HMI Dashboard",
style='Title.TLabel').pack(side='left', padx=10, pady=8)

self.conn_label = ttk.Label(title_frame, text="DISCONNECTED",
style='Title.TLabel', foreground='#f85149')

self.conn_label.pack(side='right', padx=20, pady=8)

```

```

# — Main Area —————

main = ttk.Frame(root, style='Frame.TFrame')
main.pack(fill='both', expand=True, padx=8, pady=4)

# Left column

left = ttk.Frame(main, style='Frame.TFrame')

```

```

left.pack(side='left', fill='both', expand=True, padx=4)

# RMS
self.add_section(left, "RMS VOLTAGES & CURRENTS")
rf = ttk.Frame(left, style='Frame.TFrame')
rf.pack(fill='x', padx=5)
self.va_rms = self.add_val(rf, "Va_rms", 0, 0)
self.vb_rms = self.add_val(rf, "Vb_rms", 1, 0)
self.vc_rms = self.add_val(rf, "Vc_rms", 2, 0)
self.ib_rms = self.add_val(rf, "Ib_rms", 0, 2)
self.ic_rms = self.add_val(rf, "Ic_rms", 1, 2)

# Frequency
self.add_section(left, "FREQUENCY (PDS: 49.5-50.5 Hz)")
ff = ttk.Frame(left, style='Frame.TFrame')
ff.pack(fill='x', padx=5)
self.f_va = self.add_val(ff, "f_Va", 0, 0)
self.f_vb = self.add_val(ff, "f_Vb", 1, 0)
self.f_vc = self.add_val(ff, "f_Vc", 2, 0)

# THD
self.add_section(left, "THD (PDS limit: 4.5% EREC G5/5)")
tf = ttk.Frame(left, style='Frame.TFrame')
tf.pack(fill='x', padx=5)
self.thd_va = self.add_val(tf, "THD_Va", 0, 0)
self.thd_vb = self.add_val(tf, "THD_Vb", 1, 0)
self.thd_vc = self.add_val(tf, "THD_Vc", 2, 0)

# Waveform
self.add_section(left, "THREE PHASE WAVEFORM (reconstructed)")

```

```

        self.wave_canvas = tk.Canvas(left, bg='#0d1117', height=130,
highlightthickness=0, bd=0)

        self.wave_canvas.pack(fill='x', padx=5, pady=3)


# Right column
right = ttk.Frame(main, style='Frame.TFrame')
right.pack(side='right', fill='both', expand=True, padx=4)


# Phase
self.add_section(right, "PHASE ANGLE")
pf = ttk.Frame(right, style='Frame.TFrame')
pf.pack(fill='x', padx=5)
self.ph_vbib = self.add_val(pf, "Ph Vb/Ib", 0, 0)
self.ph_vcic = self.add_val(pf, "Ph Vc/Ic", 1, 0)


# Power
self.add_section(right, "POWER QUANTITIES")
pwf = ttk.Frame(right, style='Frame.TFrame')
pwf.pack(fill='x', padx=5)
self.pb = self.add_val(pwf, "Pb", 0, 0)
self.qb = self.add_val(pwf, "Qb", 0, 2)
self.pfb_v = self.add_val(pwf, "PFb", 0, 4)
self.pc = self.add_val(pwf, "Pc", 1, 0)
self.qc = self.add_val(pwf, "Qc", 1, 2)
self.pfc_v = self.add_val(pwf, "PFc", 1, 4)


# Alarms
self.add_section(right, "ALARM STATUS")
af = ttk.Frame(right, style='Frame.TFrame')
af.pack(fill='x', padx=5, pady=3)

```

```

        self.alarm_freq = ttk.Label(af, text="      FREQ:  OK      ",
style='AlarmOK.TLabel')

        self.alarm_freq.grid(row=0, column=0, padx=3, pady=3)

        self.alarm_uv = ttk.Label(af, text="      UV:  OK      ",
style='AlarmOK.TLabel')

        self.alarm_uv.grid(row=0, column=1, padx=3, pady=3)

        self.alarm_thd = ttk.Label(af, text="      THD:  OK      ",
style='AlarmOK.TLabel')

        self.alarm_thd.grid(row=0, column=2, padx=3, pady=3)


# Compliance
self.add_section(right, "PDS COMPLIANCE")
cf = ttk.Frame(right, style='Frame.TFrame')
cf.pack(fill='x', padx=5)
self.comp_freq = self.add_comp(cf, "Frequency", 0)
self.comp_thd = self.add_comp(cf, "THD", 1)
self.comp_pf = self.add_comp(cf, "Power Factor", 2)


# DNP3
self.add_section(right, "DNP3 IEEE 1815 TELEMETRY")
df = ttk.Frame(right, style='Frame.TFrame')
df.pack(fill='x', padx=5)

self.dnp3_info = ttk.Label(df, text="Waiting for frames...",
style='Status.TLabel')

self.dnp3_info.pack(anchor='w')

self.dnp3_hex = tk.Text(df, height=2, width=55, bg='#0a0a1a',
fg='#58a6ff', font=('Courier', 7), state='disabled', wrap='word')

self.dnp3_hex.pack(fill='x', pady=2)


# Status bar
sb = tk.Frame(root, bg='#161b22', height=25)
sb.pack(fill='x', side='bottom')

```

```

        sb.pack_propagate(False)

        self.status = ttk.Label(sb, text="Starting...",
style='Status.TLabel', background='#161b22')

        self.status.pack(side='left', padx=10, pady=3)

    def update_gui():

    def add_section(self, parent, title):

        tk.Frame(parent, bg='#30363d', height=1).pack(fill='x', padx=5,
pady=(8, 2))

        ttk.Label(parent, text=title, style='Header.TLabel').pack(anchor='w',
padx=5)

    def add_val(self, parent, label, row, col):

        ttk.Label(parent, text=f"{label}:",
style='Label.TLabel').grid(row=row, column=col, sticky='w', padx=(5,2),
pady=1)

        v = ttk.Label(parent, text="---", style='Value.TLabel')

        v.grid(row=row, column=col+1, sticky='w', padx=(2,10), pady=1)

        return v

    def add_comp(self, parent, label, row):

        ttk.Label(parent, text=f"{label}:",
style='Label.TLabel').grid(row=row, column=0, sticky='w', padx=(5,2), pady=1)

        v = ttk.Label(parent, text="---", style='Compliance.TLabel')

        v.grid(row=row, column=1, sticky='w', padx=(2,10), pady=1)

        return v

    def set_alarm(self, widget, name, bad):

        if bad:

            widget.configure(text=f"
{name}: ALARM
",
style='AlarmBAD.TLabel')

```

```

else:
    widget.configure(text=f" {name}: OK ", style='AlarmOK.TLabel')

def draw_waveform(self):
    c = self.wave_canvas
    c.delete('all')
    w = c.winfo_width()
    h = c.winfo_height()
    if w < 10 or h < 10:
        return

    mid = h // 2
    # Grid
    c.create_line(0, mid, w, mid, fill='#30363d')
    c.create_line(0, mid - h//4, w, mid - h//4, fill='#21262d', dash=(2,4))
    c.create_line(0, mid + h//4, w, mid + h//4, fill='#21262d', dash=(2,4))

    d = self.data
    amp_a = max(d.Va_rms * 1.414, 0.01)
    amp_b = max(d.Vb_rms * 1.414, 0.01)
    amp_c = max(d.Vc_rms * 1.414, 0.01)
    max_amp = max(amp_a, amp_b, amp_c, 0.1)
    scale = (h * 0.4) / max_amp

    # Draw 3 cycles
    pts_a, pts_b, pts_c = [], [], []
    for x in range(w):
        t = x / w * 3 * 2 * math.pi + self.wave_phase
        ya = mid - int(amp_a * math.sin(t) * scale)
        yb = mid - int(amp_b * math.sin(t - 2*math.pi/3) * scale)

```

```

        yc = mid - int(amp_c * math.sin(t - 4*math.pi/3) * scale)
        pts_a.extend([x, ya])
        pts_b.extend([x, yb])
        pts_c.extend([x, yc])

    if len(pts_a) > 3:
        c.create_line(*pts_a, fill='#f85149', width=2, smooth=True)
        c.create_line(*pts_b, fill='#3fb950', width=2, smooth=True)
        c.create_line(*pts_c, fill='#58a6ff', width=2, smooth=True)

    # Legend
    c.create_text(w-80, 12, text="Va", fill='#f85149', font=('Consolas',
8, 'bold'), anchor='w')
    c.create_text(w-55, 12, text="Vb", fill='#3fb950', font=('Consolas',
8, 'bold'), anchor='w')
    c.create_text(w-30, 12, text="Vc", fill='#58a6ff', font=('Consolas',
8, 'bold'), anchor='w')

    self.wave_phase += 0.15

def update_gui(self):
    d = self.data

    # Connection
    if d.connected:
        self.conn_label.configure(text="CONNECTED", foreground='#3fb950')
    else:
        self.conn_label.configure(text="DISCONNECTED",
foreground='#f85149')

    # RMS

```



```
self.va_rms.configure(text=f"{d.Va_rms:.4f} V")
self.vb_rms.configure(text=f"{d.Vb_rms:.4f} V")
self.vc_rms.configure(text=f"{d.Vc_rms:.4f} V")
self.ib_rms.configure(text=f"{d.Ib_rms:.4f} V")
self.ic_rms.configure(text=f"{d.Ic_rms:.4f} V")

# Frequency
self.f_va.configure(text=f"{d.f_Va:.4f} Hz")
self.f_vb.configure(text=f"{d.f_Vb:.4f} Hz")
self.f_vc.configure(text=f"{d.f_Vc:.4f} Hz")

# THD
self.thd_va.configure(text=f"{d.THD_Va:.2f} %")
self.thd_vb.configure(text=f"{d.THD_Vb:.2f} %")
self.thd_vc.configure(text=f"{d.THD_Vc:.2f} %")

# Phase
self.ph_vbib.configure(text=f"{d.Ph_VbIb:.2f} deg")
self.ph_vcic.configure(text=f"{d.Ph_VcIc:.2f} deg")

# Power
self.pb.configure(text=f"{d.Pb:.4f}")
self.qb.configure(text=f"{d.Qb:.4f}")
self.pfb_v.configure(text=f"{d.PFb:.3f}")
self.pc.configure(text=f"{d.Pc:.4f}")
self.qc.configure(text=f"{d.Qc:.4f}")
self.pfc_v.configure(text=f"{d.PFc:.3f}")

# Alarms
self.set_alarm(self.alarm_freq, "FREQ", d.freq_alarm)
```

```

self.set_alarm(self.alarm_uv, "UV", d.uv_alarm)
self.set_alarm(self.alarm_thd, "THD", d.thd_alarm)

# Compliance
freq_ok = all(49.5 <= f <= 50.5 for f in [d.f_Va, d.f_Vb, d.f_Vc] if f
> 0)

thd_ok = all(t < 4.5 for t in [d.THD_Va, d.THD_Vb, d.THD_Vc])
pf_ok = all(p > 0.5 for p in [d.PFb, d.PFc] if p > 0)

self.comp_freq.configure(
    text="PASS 49.5-50.5Hz" if freq_ok else "FAIL",
    foreground='#3fb950' if freq_ok else '#f85149')
self.comp_thd.configure(
    text="PASS <4.5% G5/5" if thd_ok else "FAIL >4.5%",
    foreground='#3fb950' if thd_ok else '#f85149')
self.comp_pf.configure(
    text=f"PASS PFb:{d.PFb:.2f} PFc:{d.PFc:.2f}" if pf_ok else
"CHECK",
    foreground='#3fb950' if pf_ok else '#d2a8ff')

# DNP3
self.dnp3_info.configure(
    text=f"Frames: {d.frame_count} | Size: {d.dnp3_bytes} bytes |
Src:0x0001 Dst:0x0003 | Last: {d.last_update}")
if d.raw_frame:
    self.dnp3_hex.configure(state='normal')
    self.dnp3_hex.delete('1.0', 'end')
    self.dnp3_hex.insert('1.0', d.raw_frame[:250])
    self.dnp3_hex.configure(state='disabled')

# Status

```

```

        if d.connected:
            self.status.configure(
                text=f"    Connected to serial    |    Frames: {d.frame_count}    |
Last: {d.last_update}    |    DNP3 IEEE 1815 Unsolicited Response",
                foreground='#3fb950')
        else:
            self.status.configure(text="        Waiting    for    connection...",
foreground='#f85149')

        # Waveform
        self.draw_waveform()

        self.root.after(500, self.update_gui)

```

— Port Selection —————

```

def select_port():
    ports = serial.tools.list_ports.comports()
    print("\nAvailable COM ports:")
    print("-" * 40)
    for i, p in enumerate(ports):
        print(f"    [{i+1}] {p.device}: {p.description}")
    print("-" * 40)

    if not ports:
        print("No COM ports found!")
        port = input("Enter COM port manually (e.g. COM6): ").strip()
        return port

```

```

# Check command line argument
if len(sys.argv) > 1:
    return sys.argv[1]

choice = input(f"Enter port number [1-{len(ports)}] or COM name: ").strip()
try:
    idx = int(choice) - 1
    if 0 <= idx < len(ports):
        return ports[idx].device
except ValueError:
    pass

if choice.upper().startswith('COM'):
    return choice.upper()

if ports:
    return ports[0].device
return 'COM6'

```

— Main —————

```

def main():
    print("=" * 50)
    print("  MIMO RTU - SCADA HMI Dashboard")
    print("  DNP3 IEEE 1815 Telemetry Receiver")
    print("=" * 50)

    port = select_port()
    print(f"\nConnecting to: {port}")

```

```
print("Close Arduino Serial Monitor first!\n")

data = RTUData()
reader = SerialReader(port, 115200, data)
reader.start()

root = tk.Tk()
app = SCADAApp(root, data)
root.mainloop()
reader.running = False

if __name__ == '__main__':
    main()
```

The dashboard executable is run from a terminal with ‘python SCADA_HMI.py’ and either auto-detects the COM port from the system port list or accepts a COM name as a command-line argument. The Tkinter UI launches once a valid serial connection is established, and the CONNECTED / DISCONNECTED status indicator in the title bar follows the port state via the SerialReader thread’s connected flag. The complete dashboard layout, with all twelve analogue parameters, three alarm tiles, the PDS compliance strip and the live DNP3 hex frame inspector, is shown in Figure 4.10.